SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL

Proposed Scheme & Syllabus

for

Master of Technology (Artificial Intelligence)

W.E.F 2025-26

School of Engineering and Technology

SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL

M Tech (Artificial Intelligence)
Program Structure
Program Educational Objectives (PEOs)

PEO-1: Students shall have the ability to apply knowledge across the disciplines and in emerging areas of Artificial Intelligence (AI) for higher studies, research, employability, product development and handle the realistic problems.

PEO-2: To understand industry careers involving innovation and programming solving using AI and Machine Learning technologies.

PEO-3: Students shall possess academic excellence with innovative insight, soft skills, managerial skills, leadership qualities, knowledge of contemporary issues and understand the need for lifelong learning for a successful professional career.

PEO-4: Students will have the ability to apply the gained knowledge to improve the society ensuring ethical and moral values.

PEO-5: Promote Design, Research, and implementation of products and services in the field of Artificial Intelligence through strong communication and entrepreneurial skills.

Program Outcomes (POs)

PO-1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO-2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO-3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO-4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO-5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO-6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO-7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO-8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO-9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO-10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO-11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one‘s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO-12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

Post Graduates of the program will be able to:

PSO1: Apply advanced knowledge of Artificial Intelligence, algorithms, and technologies to design and develop innovative AI-based solutions for real-world problems.

PSO2: Analyze, design, and implement intelligent systems by applying higher-order thinking to solve complex problems across diverse domains using AI and Data Science tools.

PSO3: Demonstrate research aptitude and an entrepreneurial mindset by identifying, formulating, and addressing contemporary issues in AI, leading to technological advancement and innovation.

PSO4: Integrate ethical values, sustainability principles, and human-centric approaches in the development and deployment of AI solutions for the betterment of society.

SANJEEV AGRAWAL GLOBAL EDUCATIONAL UNIVERSITY, BHOPAL

SCHOOL OF ENGINEERING & TECHNOLOGY

SEMESTER- I

S. No.	Course Code	Category	Course Title	Maximum Marks Allotted										Credits	Credits Hours
				Theory (100 Marks)			Studio/Practical (50 marks)			Total	Hours / Week				
				End Sem. Exam (ESE)	Continuous Internal Evaluation (CIE)		End Sem. Exam (ESE) For Practical and Viva	Continuous Internal Evaluation (CIE)		Total Marks	L	T	P		
					Mid Sem. Test (MST)	Attendance/ Quiz/ Assignment		Lab Work / WPR	Attendance / Assignment / Viva/Lab Manual						
1	ET25MTAI101T	DC	Mathematics for Computing	60	20	20	-	-	-	100	4	0	0	4	4
2	ET25MTAI102T	DC	Computability, Algorithms, and Complexity	60	20	20	-	-	-	100	3	0	0	3	3
3	ET25MTAI103T	DC	Artificial Intelligence & Foundation of ML	60	20	20	-	-	-	100	3	0	0	3	3
4	To be Offered **	DE	Departmental Elective - I	60	20	20	-	-	-	100	3	0	0	3	3
5	To be Offered **	DE	Departmental Elective - II	60	20	20	-	-	-	100	4	0	0	4	4
6	ET25MTAI102P	DC	Computability, Algorithms, and Complexity Lab	-	-	-	30	10	10	50	0	0	2	1	2
7	ET25MTAI103P	DC	Artificial Intelligence & Foundation of ML Lab	-	-	-	30	10	10	50	0	0	2	1	2
8	To be Offered **	DSEEC	Departmental Skill Enhancement Elective Course - I	-	-	-	30	10	10	50	0	0	4	2	4
9	To be Offered **	DE	Departmental Elective - I Lab	-	-	-	30	10	10	50	0	0	2	1	2
Total:				**300**	**100**	**100**	**120**	**40**	**40**	**700**	**17**	**0**	**10**	**22**	**27**

SANJEEV AGRAWAL GLOBAL EDUCATIONAL UNIVERSITY, BHOPAL
SCHOOL OF ENGINEERING & TECHNOLOGY
SEMESTER- II

S. No.	Course Code	Category	Course Title	Maximum Marks Allotted										Credits	Credits Hours
				Theory (100 Marks)			Studio/Practical (50 marks)			Total	Hours / Week				
				End Sem. Exam (ESE)	Continuous Internal Evaluation (CIE)		End Sem. Exam (ESE) For Practical and Viva	Continuous Internal Evaluation (CIE)		Total Marks	L	T	P		
					Mid Sem. Test (MST)	Attendance/ Quiz/ Assignment		Lab Work / WPR	Attendance / Assignment / Viva/Lab Manual						
1	ET25MTAI201T	DC	Computational Perception	60	20	20	-	-	-	100	4	0	0	4	4
2	ET25MTAI202T	DC	Deep Learning	60	20	20	-	-	-	100	3	0	0	3	3
3	To be Offered **	DE	Departmental Elective - III	60	20	20	-	-	-	100	3	0	0	3	3
4	To be Offered **	DE	Departmental Elective - IV	60	20	20	-	-	-	100	4	0	0	4	4
5	To be Offered **	DE	Departmental Elective - V	60	20	20	-	-	-	100	4	0	0	4	4
6	ET25MTAI202P	DC	Deep Learning Lab	-	-	-	30	10	10	50	0	0	2	1	2
7	To be Offered **	DE	Departmental Elective - III Lab	-	-	-	30	10	10	50	0	0	2	1	2
8	To be Offered **	DSEEC	Departmental Skill Enhancement Elective Course - II	-	-	-	30	10	10	50	0	0	4	2	4
Total:				**300**	**100**	**100**	**90**	**30**	**30**	**650**	**18**	**0**	**8**	**22**	**26**

SANJEEV AGRAWAL GLOBAL EDUCATIONAL UNIVERSITY, BHOPAL
SCHOOL OF ENGINEERING & TECHNOLOGY
SEMESTER- III

S. No.	Course Code	Category	Course Title	Maximum Marks Allotted											Credits	Credits Hours
				Theory (100 Marks)			Studio/Practical (50 marks)			Total	Hours / Week					
				End Sem. Exam (ESE)	Continuous Internal Evaluation (CIE)		End Sem. Exam (ESE) For Practical and Viva	Continuous Internal Evaluation (CIE)		Total Marks	L	T	P			
					Mid Sem. Test (MST)	Attendance/ Quiz/ Assignment		Lab Work / WPR	Attendance / Assignment / Viva/Lab Manual							
1	ET25MTAI301T	DC	Data Mining & Predictive Modelling	60	20	20	-	-	-	100	4	0	0		4	4
2	ET25MTRM302T	DC	Research Methodology	60	20	20	-	-	-	100	4	0	0		4	4
3	To be Offered **	DE	Departmental Elective - VI	60	20	20	-	-	-	100	4	0	0		4	4
4	ET25MTAI303P	PR	Dissertation Phase - I	-	-	-	100	40	60	100	0	0	12		6	
5	ET25MTAI304P	DS	Industrial Internship	-	-	-	50	20	30	100	0	0	4		2	
Total:				**180**	**60**	**60**	**150**	**60**	**90**	**500**	**12**	**0**	**16**		**20**	**12**

SANJEEV AGRAWAL GLOBAL EDUCATIONAL UNIVERSITY, BHOPAL

SCHOOL OF ENGINEERING & TECHNOLOGY

SEMESTER- IV

<table>
<tr><th rowspan="4">S. No.</th><th rowspan="4">Course Code</th><th rowspan="4">Category</th><th rowspan="4">Course Title</th><th colspan="10">Maximum Marks Allotted</th><th rowspan="4">Credits</th><th rowspan="4">Credits Hours</th></tr>
<tr><th colspan="3">Theory (100 Marks)</th><th colspan="3">Studio/Practical (50 marks)</th><th>Total</th><th colspan="3">Hours / Week</th></tr>
<tr><th rowspan="2">End Sem. Exam (ESE)</th><th colspan="2">Continuous Internal Evaluation (CIE)</th><th rowspan="2">End Sem. Exam (ESE) For Practical and Viva</th><th colspan="2">Continuous Internal Evaluation (CIE)</th><th rowspan="2">Total Marks</th><th rowspan="2">L</th><th rowspan="2">T</th><th rowspan="2">P</th></tr>
<tr><th>Mid Sem. Test (MST)</th><th>Attendance/ Quiz/ Assignment</th><th>Lab Work / WPR</th><th>Attendance / Assignment / Viva/Lab Manual</th></tr>
<tr><td>1</td><td>ET25MTAI401P</td><td>PR</td><td>Dissertation Phase II</td><td>-</td><td>-</td><td>-</td><td>200</td><td>100</td><td>100</td><td>400</td><td>0</td><td>0</td><td>30</td><td>15</td><td>0</td></tr>
<tr><td colspan="4">Total:</td><td>-</td><td>-</td><td>-</td><td>200</td><td>100</td><td>100</td><td>400</td><td>0</td><td>0</td><td>30</td><td>15</td><td>0</td></tr>
</table>

Curriculum Components

Components	Credits
Program Core (10Courses)	**28**
Departmental Elective (08Courses)	**24**
Departmental Skill Enhancement Elective Course (05Courses)	**27**
Total	**79**

Distribution of credits across all components

SEM	Departmental Core	Departmental Elective	Departmental Skill Enhancement Elective Course	Total Credit
I.	5	3	1	**22**
II.	3	4	1	**22**
III.	2	1	2	**20**
IV.	0	0	1	**15**
Total	**10**	**8**	**5**	**79**

Table-1
List of Departmental Skill Enhancement Elective Course (DSEEC)

SN	Course Code	DSEEC -I
1.	ET25MTAI115P	Programming in PYTHON
2.	ET25MTAI116P	Programming in Java
3.	ET25MTAI117P	LISP Programming
SN	**Course Code**	**DSEEC -II**
1.	ET25MTAI217P	Advance DBMS
2.	ET25MTAI218P	Java Scripting with AJAX

Table-2
List of Departmental Elective (DE)

SN	Course Code	DE-I
1.	ET25MTAI111T/P	Artificial Neural Networks
2.	ET25MTAI112T/P	Introduction to Problem Solving & Reasoning
SN	**Course Code**	**DE-II**
1.	ET25MTAI113T	Information Security & Cyber Law
2.	ET25MTAI114T	Evolutionary Computing
SN	**Course Code**	**DE-III**
1.	ET25MTAI211T/P	Artificial Intelligence for Robotics
2.	ET25MTAI212T/P	Digital Image Processing
SN	**Course Code**	**DE-IV**
1.	ET25MTAI213T	Deductive Systems
2.	ET25MTAI214T	Decision Making Under Uncertainty
SN	**Course Code**	**DE-V**
1.	ET25MTAI215T	Data privacy and user protection
2.	ET25MTAI216T	Information Retrieval
SN	**Course Code**	**DE-VI**
1.	ET25MTCS113T	Natural Language Processing
2.	ET25MTAI312T	Computer Vision

SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL

Proposed Scheme & Syllabus

for

Master of Technology (Artificial Intelligence)

I Semester

School of Engineering and Technology

<table>
<tr><td colspan="8">SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL</td></tr>
<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="5">Department Name: Advance Computing</td></tr>
<tr><td colspan="6">Recommended Programs: M. Tech. in Artificial Intelligence</td><td colspan="2">Semester: I</td></tr>
<tr><td>Course Name</td><td colspan="2">Mathematics for Computing</td><td>Course Code</td><td colspan="4">ET25MTAI101T</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2">Total Credits</td><td colspan="2" rowspan="2">4</td></tr>
<tr><td>4</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="7">Mathematics of Class 11th & 12th</td></tr>
<tr><td>Course Objectives</td><td colspan="7">1.To be able to understand advanced multivariate statistical techniques and Automaton Techniques.
2.To understand the factor analysis and estimation.
3.To understand the production systems and their grammar.
4.Discussion of Finite-state Automata and Nondeterministic FSA.
5.To learn the construction of Turing Machines.</td></tr>
<tr><td rowspan="5">Course Content</td><td colspan="6">Unit 1:
Advanced Multivariate Statistical Techniques
Multiple linear regression, multicollinearity, outliers, heteroscedasticity, autocorrelation and model validation, polynomial regression, and the method of orthogonal polynomials, nonlinear regression, multivariate regression model, testing and estimation, Ridge regression.</td><td>12 Hours</td></tr>
<tr><td colspan="6">Unit 2:
Multivariate analysis of variance and covariance, Test of linear hypothesis, likelihood ratio criteria, a test of equality of means of several normal distributions, testing independence of a set of variates, factor analysis, extracting common factors, factor scores and estimation.</td><td>12 Hours</td></tr>
<tr><td colspan="6">Unit 3:
Cluster analysis, correlations, and distances, methods of clustering: hierarchical clustering, non-hierarchical clustering, partitioning methods, overlapping clustering
Grammars - Production systems –Chomskian Hierarchy - Right linear grammar and Finite state automata - Context free grammars - Normal forms - uvwxy theorem – Parikh mapping - Self-embedding property - Subfamilies of CFL - Derivation trees and ambiguity.</td><td>12 Hours</td></tr>
<tr><td colspan="6">Unit 4:
Finite-state Automata - Nondeterministic and deterministic FSA, NFSA with ε-moves, Regular Expressions - Equivalence of regular expression and FSA Pumping lemma, closure properties, and decidability. Myhill - Nerode theorem and minimization - Finite automata with output.</td><td>12 Hours</td></tr>
<tr><td colspan="6">Unit 5:
Pushdown automata - Acceptance by empty store and final state - Equivalence between pushdown automata and context-free grammars - Closure properties of CFL - Deterministic pushdown automata.
Turing Machines - Techniques for Turing machine construction - Generalized and restricted versions equivalent to the basic model - Godel numbering - Universal Turing Machine - Recursively enumerable sets and recursive sets - Computable functions - time-space complexity measures - context-sensitive languages and linear bound automata.</td><td>12 Hours</td></tr>
</table>

Textbooks	T1: J.E.Hopcroft,R.Motwani and J.D.Ullman , "Introduction to Automata Theory Languages and computation", Pearson Education Asia , 2001. T2: Peter Linz, "An Introduction to Formal Language and Automata", 4th Edition,Narosa Publishing house , 2006
References	R1: Kohavi, Z. Switching & Finite Automata Theory - Tata Mcgraw- Hill. R2: J.E.Hopcroft,R.Motwani and J.D.Ullman , "Introduction to Automata Theory Languages and computation", Pearson Education Asia , 2001.
Course Outcomes	**After completing this course, students will be able to:** CO 1: To be able to understand advanced multivariate statistical techniques and Automaton Techniques. CO 2: To understand the factor analysis and estimation. CO 3: To understand the production systems and their grammar. CO 4: Discussion of Finite state Automata and Nondeterministic FSA. CO 5: To learn the construction of Turing Machines.

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, ..., K_6)
CO1	2	2	2									1	1	1	**K_1, K_2, K_3**
CO2	2	2	2	1							2	3	2	2	**K_1, K_2, K_3, K_4**
CO3	3	3	2	1	3						2	3	3	2	**K_1, K_2, K_3, K_4**
CO4	3	3	2	1	3						2	3	3	2	**K_1, K_2, K_3, K_4**
CO5	3	3	3	1	3						2	3	3	2	**K_1, K_2, K_3, K_4, K_5**

High-3 Medium-2 Low-1

K_1 =>Remember K_2 =>Understand K_3 =>Apply K_4 =>Analyze K_5 =>Evaluate

K_6 =>Create

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<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="4">Department Name: Advance Computing</td></tr>
<tr><td colspan="6">Recommended Programs: M. Tech. in Artificial Intelligence</td><td>Semester: I</td></tr>
<tr><td>Course Name</td><td colspan="2">Computability, Algorithms and Complexity</td><td>Course Code</td><td colspan="3">ET25MTAI102T</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2">Total Credits</td><td rowspan="2">3</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="6">● Fundamentals of computing
● Fundamentals of programming
● Data structure & complexity</td></tr>
<tr><td>Course Objectives</td><td colspan="6">1. To identify and implement an appropriate algorithm for solving a novel problem.
2. To Analyze an algorithm’s asymptotic running time and memory consumption.
3. To Evaluate a set of proposed algorithmic solutions to a problem according to given constraints
4. To demonstrate the correctness of an algorithm.
5. To develop an appreciation and understanding of the theoretical framework for computer science and some of the consequences of this theory.</td></tr>
<tr><td rowspan="5">Course Content</td><td colspan="5">Unit- I
Computability- algorithm, Turing machines, Turing machines importance’s, Church’s thesis, Turing machines and examples , Input and output of a Turing machine, Representing Turing , Examples of Turing machines , Variants of Turing machines, Computational power , Two-way-tape Turing machines , Multi-tape Turing machines , Codes for standard Turing machines, The universal Turing machine</td><td>10 Hours</td></tr>
<tr><td colspan="5">Unit-2
Unsolvable problems – Introduction, The halting problem, Reduction, Godel’s incompleteness theorem, Part I in a nutshell</td><td>09 Hours</td></tr>
<tr><td colspan="5">Unit-3
Algorithms - Use of algorithms, Run time function of an algorithm, Choice of algorithm, Implementation</td><td>08 Hours</td></tr>
<tr><td colspan="5">Unit-4
Graph algorithms - Graphs: the basics, Representing graphs, Algorithm for searching a graph, Paths and connectedness , Trees, spanning trees , Complete graphs, Hamiltonian circuit problem (HCP), Weighted graphs, Minimal spanning trees , Prim’s algorithm to find a MST, Shortest path , Travelling salesman problem (TSP),polynomial time , NP-completeness.</td><td>08 Hours</td></tr>
<tr><td colspan="5">Unit-5
Complexity - Basic complexity theory, Yes/no problems, Polynomial-time Turing machines, Tractable problems, the class P, Intractable problems? An exhaustive search in algorithms, deterministic Turing machines, Definition of non-deterministic TM, examples of NDTMs, speed of NDTMs, and The class NP.
NP-completeness - Introduction, Proving NP-completeness by reduction, Cook’s theorem</td><td>10 Hours</td></tr>
<tr><td>Text Books</td><td colspan="6">T1: Introduction to the Theory of Computation, Sipser
T2: Computational Complexity, Papadimitriou</td></tr>
<tr><td>References</td><td colspan="6">R1: Algorithm Design, Kleinberg &Tardos
R2: Introduction to Algorithms, Cormen, Leiserson, Rivest, and Stein</td></tr>
</table>

Course Outcomes	**After completing this course, students will be able to:** CO1.The student will be able to identify and implement an appropriate algorithm for solving a novel problem; CO2.Student Will learn and analyze an algorithm's asymptotic running time and memory consumption. CO3.Students can able to Evaluate a set of proposed algorithmic solutions to a problem according to given constraints CO4.Students can demonstrate the correctness of an algorithm. CO5.To develop an appreciation and understanding of the theoretical framework for computer science and some of the consequences of this theory

Mapping of Course outcomes with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, …, K_6)
CO1	2	2	2									1	1	1	**K_1, K_2, K_3**
CO2	2	2	2	1							2	3	2	2	**K_1, K_2, K_3, K_4**
CO3	3	3	2	1	3						2	3	3	2	**K_1, K_2, K_3, K_4**
CO4	3	3	2	1	3						2	3	3	2	**K_1, K_2, K_3, K_4**
CO5	3	3	3	1	3						2	3	3	2	**K_1, K_2, K_3, K_4, K_5**

High-3 Medium-2 Low-1

K_1 =>Remember K_2 =>Understand K_3 =>Apply K_4 =>Analyze K_5 =>Evaluate

K_6 =>Create

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<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="4">Department Name: Institute of Advance Computing</td></tr>
<tr><td colspan="5">Recommended Programs: M. Tech. in Artificial Intelligence</td><td colspan="2">Semester: I</td></tr>
<tr><td>Course Name</td><td colspan="2">Artificial Intelligence & Foundation of ML</td><td>Course Code</td><td colspan="3">ET25MTAI103T</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2">Total Credits</td><td rowspan="2">3</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>-</td></tr>
<tr><td>Prerequisites (if any)</td><td colspan="6">● Fundamentals of computing
● Fundamentals of programming</td></tr>
<tr><td>Course Objectives</td><td colspan="6">1. To understand basic principles of Artificial Intelligence
2. To learn and design intelligent agents
3. To understand the concepts of artificial intelligence including problem-solving, knowledge representation, reasoning, decision making, planning, perception, and action.
4. To master the fundamentals of machine learning, mathematical framework, and learning algorithm.
5. Understand concepts of Spark, Zookeeper, and HBase</td></tr>
<tr><td rowspan="5">Course Content</td><td colspan="5">Unit I
Automated Reasoning - foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time, and space; predicate logic, situation calculus, description logics, - Logic - Propositional and predicate logic - Syntax - Informal and formal semantics - Equivalence - De Morgans laws</td><td>10 Hours</td></tr>
<tr><td colspan="5">Unit II
Decidable problems - Many-sorted logic -first-order, higher-order logic Reasoning methods - Formal program techniques - pre-and post conditions, derivation and verification of programs- SPIN Tool.</td><td>10 Hours</td></tr>
<tr><td colspan="5">Unit III
Uncertain Knowledge-yesian networks; Basics of decision theory, sequential decision problems, Elementary game theory; Problem-solving through Search-forward and backward, state-space, blind, heuristic, problem-reduction, A, A*, AO*, minimax, constraint propagation, neural and stochastic.</td><td>09 Hours</td></tr>
<tr><td colspan="5">UNIT-IV
Introduction to intelligent agents; Machine Learning - Foundations of supervised learning – Decision trees and inductive bias, Regression Vs Classification, Supervised-Linear Regression, Logistic</td><td>08 Hours</td></tr>
<tr><td colspan="5">Unit V
Generalization, Training, Validation and Testing, Problem of Overfitting, Bias vs Variance, Confusion Matrix, Precision, Recall, F Measure, Support Vector Machine, Decision Tree, Random Forest, Perceptron, Beyond binary classification, Boosting and bagging, bootstrapping – Advanced supervised learning - K-Nearest Neighbour, Markov model, Hidden Markov Model - Nearest Neighbor Classification- Gaussianprocesses UnsupervisedLearning-DimensionalityReductionTechniques, Linear Discriminant Analysis - Clustering: K-means, Hierarchical, Spectral, subspace clustering, association rule mining.</td><td>08 Hours</td></tr>
<tr><td>Textbooks</td><td colspan="6">T1: Hadoop: The Definitive Guide, 4th Edition by Tom White - Shroff Publishers & Distributers Private Limited - Mumbai; Fourth edition (2015)</td></tr>
</table>

	T2: Basics of Artificial Intelligence & Machine Learning, Dr. Dheeraj Mehrotra
References	R1: Big Data: Principles and Best Practices of Scalable Realtime Data Systems by James Warren and Nathan Marz, Manning Publications (2015) R2: Kenneth A. Lambert, "Fundamentals of Python – First Programs", CENGAGE Publication
Course Outcomes	**After completing this course, students will be able to:** CO1. Understand the concepts of Hadoop and HDFS CO2. Understand the concepts of MapReduce CO3. Analyze big data tools Pig and Hive. CO4. Understand concepts of Spark, Zookeeper and HBase CO5. Perform load balancing and serialization in Hadoop.

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, ..., K_6)
CO1	2	2	2							1	1	1	1	1	K_1, K_2, K_3
CO2	2	2	2	1						1	2	3	2	2	K_1, K_2, K_3, K_4
CO3	3	3	2	1	3					1	2	3	3	2	K_1, K_2, K_3, K_4
CO4	3	3	2	1	3					2	2	3	3	2	K_1, K_2, K_3, K_4
CO5	3	3	3	1	3					2	2	3	3	2	K_1, K_2, K_3, K_4, K_5

High-3   Medium-2   Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

K_6 => Create

SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL

School Name: School of Engineering & Technology			**Department Name: Advance Computing**			
Recommended Programs: M. Tech. in Artificial Intelligence				Semester: I		
Course Name	**Artificial Neural Networks**		**Course Code**	**ET25MTAI111T/P**		
Credit Hours	**L**	**T**	**P**	**N**	**Total Credits**	3
	3	0	0	0		
Prerequisites	• Fundamentals of computing • Fundamentals of programming • Introduction to AI & ML					
Course Objectives	1. Survey of attractive applications of Artificial Neural Networks. 2. Practical approach for using Artificial Neural Networks in various technical, organizational, and economic applications 3. Supervised learning of various knowledge representation methods 4. Unsupervised learning Probabilistic reasoning and uncertainty 5. Learn auto memory networks					
Course Content	**UNIT I:** INTRODUCTION: History of Neural Networks, Structure and Functions Of Biological And Artificial Neuron, Neural Network Architectures, Characteristics Of ANN, Basic Learning Laws and Methods.				**10 Hours**	
	UNIT II: SUPERVISED LEARNING: Single Layer Neural Network and architecture, McCulloch-Pitts Neuron Model, Learning Rules, Perceptron Model, Perceptron Convergence Theorem, Delta learning rule, ADALINE, Multi-Layer Neural Network and architecture, MADALINE, Back Propagation learning, Back Propagation Algorithm.				**10 Hours**	
	UNIT III: UNSUPERVISED LEARNING-1: Outstar Learning, Kohenen Self Organization Networks, Hamming Network And MAXNET, Learning Vector Quantization, Mexican hat.				**08 Hours**	
	UNIT IV: UNSUPERVISED LEARNING-2: Counter Propagation Network -Full Counter Propagation network, Forward Only Counter Propagation Network, Adaptive Resonance Theory (ART) -Architecture, Algorithms.				**08 Hours**	
	UNIT V: ASSOCIATIVE MEMORY NETWORKS: Introduction, Auto Associative Memory, Hetero Associative Memory, Bidirectional Associative Memory (BAM) -Theory and Architecture, BAM Training Algorithm, Hopfield Network: Introduction, Architecture Of Hopfield Network.				**09 Hours**	
Textbooks	T1: S.Rajasekaran and G.A.Vijayalakshmi pai “Neural Networks. Fuzzy Logic and Genetic Algorithms”. T2: Introduction to Artificial Neural Systems by Jacek M. Zurada					
References	R1: James A Freeman and Davis Skapura” Neural Networks Algorithm, applications, and programming Techniques”, Pearson Education, 2002. R2. Simon Hakins “Neural Networks “Pearson Education.					

Course Outcomes	**After completing this course, students will be able to:** **CO1** Students will learn about AI and Neural Networks **CO2** Students will learn about supervised learning various knowledge representation methods **CO3** Students will learn about unsupervised learning Probabilistic reasoning and uncertainty **CO4** Students will learn about auto memory networks **CO5** Learn practical approach for using Artificial Neural Networks in various technical, organizational, and economic applications

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, …, K_6)
CO1	2	2	2								1	1	1	1	**K_1, K_2, K_3**
CO2	2	2	2	1							1	3	2	2	**K_1, K_2, K_3, K_4**
CO3	3	3	2	1	3						1	3	3	2	**K_1, K_2, K_3, K_4**
CO4	3	3	2	1	3						2	3	3	2	**K_1, K_2, K_3, K_4**
CO5	3	3	3	1	3						2	3	3	2	**K_1, K_2, K_3, K_4, K_5**

High-3 **Medium-2** **Low-1**

K_1 => Remember **K_2 => Understand** **K_3 => Apply** **K_4 => Analyze** **K_5 => Evaluate**

K_6 => Create

<table>
<tr><td colspan="8">SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL</td></tr>
<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="5">Department Name: Advance Computing</td></tr>
<tr><td colspan="6">Recommended Programs: M. Tech. in Artificial Intelligence</td><td colspan="2">Semester: I</td></tr>
<tr><td>Course Name</td><td colspan="2">Introduction to Problem Solving and Reasoning</td><td>Course Code</td><td colspan="4">ET25MTAI112T/P</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2" colspan="2">Total Credits</td><td rowspan="2">3</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="7">● Fundamentals of computing
● Fundamentals of programming</td></tr>
<tr><td>Course Objectives</td><td colspan="7">1. Develop reasoning curiosity and use inductive and deductive reasoning when solving problems
2. Become confident in using reasoning to analyze and solve problems both in school and in real-life situations
3. Develop the knowledge, skills, and attitudes necessary to pursue further studies in reasoning
4. Develop abstract, logical, and critical thinking and the ability to reflect critically upon their work and the work of others
5. Develop a critical appreciation of the use of information and communication technology in reasoning</td></tr>
<tr><td rowspan="3">Course Content</td><td colspan="6">Unit I: Python Basic:
Basic syntax, running a script on jupyter, the concept of data types, variables, assignments, immutable variables, numerical types, arithmetic operators and expressions, comments in the program, understanding error messages, Conditions, boolean logic, logical operators, ranges, Control statements: if-else, loops (for, while), short-circuit (lazy) evaluation, Strings and text files, Lists, tuples, and dictionaries; basic list operators, replacing, inserting, removing an element, searching and sorting lists, dictionary literals, adding and removing keys, accessing and replacing values, traversing dictionaries. Pandas, numpy, matplotlib.</td><td>10 Hours</td></tr>
<tr><td colspan="6">Unit II: Introduction to Artificial Intelligence:
Artificial Intelligence, Programming with and without AI, Application of AI, Turing Test, Environments Constraints, agents and Environments, Environments properties, PEAS Description of task Environments, PAGE Descriptions, Example: VacBot with PEAS Description, Agent, Reflex Agent Diagram, Reflex Agent Program, Model-Based Reflex, Agent Diagram, Model-Based Reflex, Agent Program, Goal-Based Agent Diagram, Utility-Based Agent Diagram, Breadth-First Search, Learning Agents, Learning Agent Diagram.</td><td>10 Hours</td></tr>
<tr><td colspan="6">Unit III: Problem Solving Techniques.
Artificial Intelligence, Programing with and without AI, Application of AI, Turing Test, Environments Constraints, agents and Environments, Environments properties, PEAS Description of task Environments, PAGE Descriptions, Example: VacBot with PEAS Description, Agent, Reflex Agent Diagram, Reflex Agent Program, Model-Based Reflex, Agent Diagram, Model-Based Reflex, Agent Program, Goal-Based Agent Diagram, Utility-Based Agent Diagram, Breadth-First Search, Learning Agents, Learning Agent Diagram.</td><td>09 Hours</td></tr>
</table>

	Unit IV: Knowledge and Reasoning: Objectives, Knowledge Representation & Reasoning, Knowledge Representation, Reasoning, Types of Methods, Knowledge Vs Belief, Types of Knowledge, Logical Agents, Knowledge-Based Agents, Propositional Logic, Propositional Logic Table, Pros and Cons of Propositional Logic, Forward Chaining, Backward Chaining, Forward Chaining Vs Backward Chaining, First-Order Logic, Pros of First-Order Logic, Pros of First-Order Logic, Logics in General, Syntax of First-Order Logic, Components of First-Order Logic, Truth in First-Order Logic, First-Order Logic Example, Classical Planning, Introduction of Classical Planning, Search Vs Planning, The Language of Planning Problems.	**08 Hours**
	Unit V: Uncertain Knowledge and Reasoning. Objectives, Uncertainty, Sources of Uncertain knowledge, Probabilistic Reasoning, Basic Probability Theory, Conditional Probability, Bayesian Rule, The Joint Probability, Bayesian Reasoning, Ranking Potentially True Hypotheses, The Prior and Conditional Probabilities.	**08 Hours**
Text Books	T1: Bratko, Programming for Artificial Intelligence. T2: Russell, S.J. Artificial Intelligence a Modern Approach - Pearson Education.	
References	R1: Rich, E. Artificial Intelligence a Modern Approach - Tata Mcgraw- Hill R2: Bratko, Programming for Artificial Intelligence.	
Course Outcomes	**After completion of this course, students can:** **CO 1.** Comprehend the basics of python and python library. **CO 2.** Comprehend the basics of Artificial Intelligence terminologies, **CO 3.** Apply skills needed to deal with problem solving and reasoning. **CO 4.** Demonstrate propositional logic and first-order logic. **CO 5.** Apply probability and reasoning in problem-solving.	

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, …, K_6)
CO 1	1	1	1	1	2						1	1	1	1	**K_1, K_2**
CO 2	2	1	1	1	1						1	2		1	**K_1, K_2**
CO 3	2	2	1	1	1					1	1	2	1		**K_1, K_2, K_3**
CO 4	3	2	2	1	2				1		2	2	1	2	**K_1, K_2, K_3, K_4**
CO 5	3	3	2	1	2			1		1	2	2	2	2	**K_1, K_2, K_3, K_4**

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

K_6 => Create

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<tr><td colspan="8">SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL</td></tr>
<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="5">Department Name: Advance Computing</td></tr>
<tr><td colspan="5">Recommended Programs: M. Tech. in Artificial Intelligence</td><td colspan="3">Semester: I</td></tr>
<tr><td>Course Name</td><td colspan="2">Information Security & Cyber Law</td><td>Course Code</td><td colspan="4">ET25MTAI113T</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>SM</td><td rowspan="2" colspan="2">Total Credits</td><td rowspan="2">4</td></tr>
<tr><td>4</td><td>0</td><td>0</td><td>4</td></tr>
<tr><td>Prerequisites</td><td colspan="7">1.Fundamentals of computing
2.Fundamentals of cyber security</td></tr>
<tr><td>Course Objectives</td><td colspan="7">Objective of this course is to:
1.Provide information to the students regarding emerging web application vulnerabilities.
2.Deliver the information regarding tools that are utilized for the exploitation of cyber security vulnerabilities.
3.Secure both clean and corrupted systems, protect personal data, secure simple computer networks, and safe Internet usage.
4.Understand key terms and concepts in cyber law, intellectual property and cybercrimes, trademarks, and domain theft.
5.Determine computer technologies, digital evidence collection, and evidentiary reporting in forensic acquisition.</td></tr>
<tr><td rowspan="4">Course Content</td><td colspan="6">UNIT 1. Information Security & IT Cyber Laws
Need of Information Security, Attributes of Information Security, Authentication, Confidentiality, Integrity, Availability, Non-Repudiation, IT acts and Cyber Laws: IT Act: Salient Feature of IT Act 2000, Legal Provisions under the Information Technology Act, Recent amendments by the IT (Amendment Act) 2008, ActSection66(A, B, C, D, E, F), ITActSection67(A, B, C)</td><td>09 Hours</td></tr>
<tr><td colspan="6">UNIT 2. Security Services, mechanism, and attacks
Access Control, Threats and Vulnerabilities, Security Attacks, Unauthorized Access, Impersonation, Denial of Service, Malicious Software, Viruses, Worms, Trojan Horses. Definitions, Types of authentication, Password Authentication, Password Vulnerabilities &Attacks: Brute Force & Dictionary Attacks. Password Policy & Discipline, Single Sign-on – Kerberos, Alternate Approaches, Biometrics: Types of Biometric Techniques: False Rejection, False Acceptance, Crossover Error Rates.</td><td>09 Hours</td></tr>
<tr><td colspan="6">UNIT 3. Physical and System Security
Function of Operating system, Types of OS (Real-time OS, Single User Single task OS, Single User-Multi tasking System, Multiuser System), Task of OS, Process, Memory Management, Device Management, Storage Management, Application Interface, User Interface, Security Weakness, Operating System, Windows Weakness, Hardening OS during Installation, Secure User Account Policy, Strong User Password Policy, Creating a list of Services and Programs running on Server, Patching Software, Hardening Windows, Selecting File System, Active Directory / Kerberos, General Installation Rules.</td><td>09 Hours</td></tr>
<tr><td colspan="6">UNIT 4. Internet and Web security
Web Servers and Browsers, HTTP, Cookies, Caching, Plug-in, ActiveX, Java, JavaScript, Secure Socket Layer (SSL), Secure Electronic Transaction (SET). E-mail Risks, Spam, E-mail Protocols, Simple Mail Transfer Protocol (SMTP),</td><td>09 Hours</td></tr>
</table>

	Post office Protocol (POP), Internet Access Message Protocol (ICMP). Secured Mail: Pretty Good Privacy (PGP), S/MIME(Secure/Multipurpose Internet Mail Extensions)	
	UNIT 5. Network Security Fundamentals Network Devices: Switches, Routers, Firewalls, VPN Concentrators, Load Balancers, Proxies. Network Protocol: Overview of IPV4 and IPV6, OSI Model, Maximum Transfer Unit, Internet Protocol (IP), Transport Control Protocol (TCP), User Datagram Protocol (UDP), Internet Control Message Protocol (ICMP), Address Resolution Protocol (ARP), Reverse Address Resolution Protocol (RARP), Domain Name System (DNS). Network Design: Network Address Translation (NAT), Demilitarized Zone (DMZ), Subnetting, Switching, Virtual Local Area Network (VLAN). Network Attack: Buffer Overflow, TCP Session, Hijacking, Sequence Guessing, Network Scanning. IP Security overview and architecture.	**09 Hours**
Text Books	T1: Gunter Ollmann 2007. The Phishing Guide Understanding & Preventing Phishing Attacks. IBM Internet Security Systems. T2: Thomas Erl, Ricardo Puttini, ZaighamMahmood, Cloud Computing: Concepts, Technology & Architecture, Prentice-Hall, 2013. T3: Rosenblatt B., Tripp B., Mooney S., "Digital Rights Management: Business and Technology", John Wiley & Sons, 2001.	
References	R1: RajkumarBuyya, Christian Vecchiola, S. ThamaraiSelvi, Mastering Cloud Computing, Tata McGraw-Hill Education, 2013. R2: M. N. Omar et al, “Hybrid Stepping Stone Detection Method,” in the proceeding of 1st IEEE Conference on Distributed Framework and Applications (DFmA – 2008), pp. 134–138, 2008 R3: J. Yang, Shou-Hsuan Stephen Huang, “Matching TCP packets and its application to the detection of long connection chains on the Internet,” in the proceeding of 19th IEEE International Conference on Advanced Information Networking and Applications (AINA), pp. 1005-1010,2005	
Course Outcomes	**After completing this course, students will be able to:** CO1 - Describe and analyze the hardware, software, components of a network and the interrelations & evaluate and compare systems software and emerging technologies. CO2 - Explain networking protocols and their hierarchical relationship with hardware and software. CO3 - Compare protocol models and select appropriate protocols for a particular design. CO4 - Manage multiple operating systems, systems software, network services, and security. CO5 - Determine computer technologies, digital evidence collection, and evidentiary reporting in forensic acquisition.	

Mapping of Course outcome with Program Outcomes, PSO’s, and Knowledge Levels (As per Blooms Taxonomy)

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	Knowledge Levels (K_1, K_2, ..., K_6)
CO1	1	1	1	1	2						1	K_1, K_2
CO2	2	1	1	1	1						1	K_1, K_2
CO3	2	2	1	1	1					1	1	K_1, $K_{2,}$ K_3
CO4	3	2	2	1	2				1		2	K_1, $K_{2,}$ $K_{3,}$$K_4$
CO5	3	3	2	1	2			1		1	2	K_1, $K_{2,}$ K_3, K_4

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate K_6 => Create

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<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="4">Department Name: Advance Computing</td></tr>
<tr><td colspan="5">Recommended Programs: M. Tech. in Artificial Intelligence</td><td colspan="2">Semester: I</td></tr>
<tr><td>Course Name</td><td colspan="2">Evolutionary Computing</td><td>Course Code</td><td colspan="3">ET25MTAI114T</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2">Total Credits</td><td rowspan="2">4</td></tr>
<tr><td>4</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="6">● Fundamentals of computing
● Fundamentals of programming</td></tr>
<tr><td>Course Objectives</td><td colspan="6">1. Understand the relations between the most important evolutionary algorithms.
2. New algorithms to be found in the literature now or in the future, and other search and optimization techniques.
3. Understand the implementation issues of evolutionary algorithms.
4. Design new evolutionary operators, representations, and fitness
5. Functions for specific practical and scientific applications.</td></tr>
<tr><td>Course Content</td><td colspan="6">Unit- I
Introduction to Evolutionary Computation
◦ Biological and artificial evolution
◦ Evolutionary computation and AI
◦ Different historical branches of EC, e.g., GAs, EP, ES, GP, etc.
◦ A simple evolutionary algorithm

Unit –II Search Operators
◦ Recombination/Crossover for strings (e.g., binary strings), e.g., one-point, multi-point, and uniform crossover operators
◦ Mutation for strings, e.g., bit-flipping
◦ Recombination/Crossover and mutation rates
◦ Recombination for real-valued representations, e.g., discrete and intermediate recombination
◦ Mutation for real-valued representations, e.g., Gaussian and Cauchy mutations, self-adaptive mutations, etc.
◦ Why and how a recombination or mutation operator works
Mixing different search operators
◦ An anomaly of self-adaptive mutations
◦ The importance of representation, e.g., binary vs. Gray coding
◦ Adaptive representations

Unit- III Selection Schemes
◦ Fitness proportional selection and fitness scaling
◦ Ranking, including linear, power, exponential and other ranking methods
◦ Tournament selection
◦ Selection pressure and its impact on evolutionary search

Evolutionary Combinatorial Optimization
◦ Evolutionary algorithms for TSPs
◦ Evolutionary algorithms for lecture room assignment
◦ Hybrid evolutionary and local search algorithms
Coévolution</td></tr>
</table>

	◦ Coopérative coévolution ◦ Compétitive coévolution **Niching and Speciation** ◦ Fitness sharing (explicit and implicit) ◦ Crowding and mating restriction **Unit-IV Constraint Handling** ◦ Common techniques, e.g., penalty methods, repair methods, etc. ◦ Analysis ◦ Some examples **Genetic Programming** ◦ Trees as individuals ◦ Major steps of genetic programming, e.g., functional and terminal sets, initialization, crossover, mutation, fitness evaluation, etc. ◦ Search operators on trees ◦ Automatically defined functions ◦ Issues in genetic programming, e.g., bloat, scalability, etc. ◦ Examples **Multi-objective Evolutionary Optimization** ◦ Pareto optimality ◦ Multiobjective evolutionary algorithms **Unit-V Learning Classifier Systems** ◦ Basic ideas and motivations ◦ Main components and the main cycle ◦ Credit assignment and two approaches **Theoretical Analysis of Evolutionary Algorithms** ◦ Schema theorems ◦ Convergence of EAs ◦ Computational time complexity of EAs ◦ No free lunch theorem
Textbooks	T1: Title: Handbook on Evolutionary Computation Author: T. Baeck, D. B. Fogel, and Z. Michalewicz (eds.) Publisher: IOP Press, 1997. T2: Title: Evolutionary Computation: Theory and Applications Author: X. Yao (ed) Publisher: World Scientific Publ. Co., Singapore, 1999. (ISBN 3-540-65907-2)
References	R1: Title: Genetic Algorithms + Data Structures = Evolution Programs (3rd edition) Author: Z Michalewicz Publisher: Springer-Verlag, Berlin, 1996 R2: Title: Evolutionary Computation: Theory and Applications Author: X. Yao (ed) Publisher: World Scientific Publ. Co. Singapore, 1999. (ISBN 3-540-65907-2)
Course Outcomes	**After completing this course, students will be able to:** CO1. Understand the relations between the most important evolutionary algorithms presented in the course, new algorithms to be found in the literature now or in the future, and other search and optimization techniques. CO2. Understand the implementation issues of evolutionary algorithms. CO3. Determine the appropriate parameter settings to make different evolutionary algorithms work well. CO4. Design new evolutionary operators, representations, and fitness CO5. Functions for specific practical and scientific applications.

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3	Knowledge Levels (K_1, K_2, ..., K_6)
CO1	2	2	2								1	1	1	1	K_1, K_2, K_3
CO2	2	2	2	1							1	3	2	2	K_1, K_2, K_3, K_4
CO3	3	3	2	1	3						1	3	3	2	K_1, K_2, K_3, K_4
CO4	3	3	2	1	3						2	3	3	2	K_1, K_2, K_3, K_4
CO5	3	3	3	1	3						2	3	3	2	K_1, K_2, K_3, K_4, K_5

High-3 **Medium-2** **Low-1**

K_1 => Remember **K_2 => Understand** **K_3 => Apply** **K_4 => Analyze** **K_5 => Evaluate**

K_6 => Create

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<tr><td colspan="8">SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL</td></tr>
<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="5">Department Name: Advance Computing</td></tr>
<tr><td colspan="6">Recommended Programs: M. Tech. in Artificial Intelligence</td><td colspan="2">Semester: I</td></tr>
<tr><td>Course Name</td><td colspan="2">Computability, Algorithms and Complexity Lab</td><td>Course Code</td><td colspan="4">ET25MTAI102P</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2">Total Credits</td><td colspan="2" rowspan="2">1</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="7">● Fundamentals of computing
● Fundamentals of programming
● Data structure & complexity</td></tr>
<tr><td>Course Objectives</td><td colspan="7">1. To identify and implement an appropriate algorithm for solving a novel problem.
2. To Analyze an algorithm’s asymptotic running time and memory consumption.
3. To Evaluate a set of proposed algorithmic solutions to a problem according to given constraints
4. To demonstrate the correctness of an algorithm.
5. To develop an appreciation and understanding of the theoretical framework for computer science and some of the consequences of this theory.</td></tr>
<tr><td>Course Content</td><td colspan="7">List of Experiments:
1. Implement Recursive Binary search and Linear search and determine the time taken to search an element
2. Sort a given set of elements using the Heap sort method and determine the time taken to sort the elements.
3. Sort a given set of elements using the Merge sort method and determine the time taken to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted, and plot a graph of the time taken versus n.
4. Sort a given set of elements using Selection sort and hence find the time required to sort elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted, and plot a graph of the time taken versus n.
5. Obtain the Topological ordering of vertices in a given digraph.
6. Implement All Pair Shortest paths problem using Floyd'sAlgorithm
7. Implement the 0/1 Knapsack problem using dynamic programming.
8. From a given vertex in a weighted connected graph, find the shortest paths to other vertices using Dijkstra's algorithm.
9. Sort a given set of elements using the Quick sort method and determine the time taken to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted, and plot a graph of the time taken versus n.
10. Find the Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm</td></tr>
<tr><td>Textbooks</td><td colspan="7">T1: Introduction to the Theory of Computation, Sipser
T2: Computational Complexity, Papadimitriou</td></tr>
<tr><td>References</td><td colspan="7">R1: Algorithm Design, Kleinberg &Tardos
R2: Introduction to Algorithms, Cormen, Leiserson, Rivest, and Stein</td></tr>
<tr><td>Course Outcomes</td><td colspan="7">After completing this course, students will be able to:
CO1.The student will be able to identify and implement an appropriate algorithm for solving a novel problem.</td></tr>
</table>

	CO2.Student Will learn and analyze an algorithm's asymptotic running time and memory consumption. CO3.Students can be able to Evaluate a set of proposed algorithmic solutions to a problem according to given constraints CO4.Students can demonstrate the correctness of an algorithm. CO5.To develop an appreciation and understanding of the theoretical framework for computer science and some of the consequences of this theory

Mapping of Course outcomes with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, …, K_6)
CO1	2	2	2									1	1	1	**K_1, K_2, K_3**
CO2	2	2	2	1							2	3	2	2	**K_1, K_2, K_3, K_4**
CO3	3	3	2	1	3						2	3	3	2	**K_1, K_2, K_3, K_4**
CO4	3	3	2	1	3						2	3	3	2	**K_1, K_2, K_3, K_4**
CO5	3	3	3	1	3						2	3	3	2	**K_1, K_2, K_3, K_4, K_5**

High-3 Medium-2 Low-1

K_1 =>Remember K_2 =>Understand K_3 =>Apply K_4 =>Analyze K_5 =>Evaluate

K_6 =>Create

SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL						
School Name: School of Engineering & Technology			**Department Name: Institute of Advance Computing**			
Recommended Programs: M. Tech. in Artificial Intelligence					**Semester: I**	
Course Name	**Artificial Intelligence & Foundation of ML Lab**		**Course Code**	**ET25MTAI103P**		
Credit Hours	**L**	**T**	**P**	**N**	**Total Credits**	1
	0	0	2	-		
Prerequisites (if any)	• Fundamentals of computing • Fundamentals of programming					
Course Objectives	1. To understand basic principles of Artificial Intelligence 2. To learn and design intelligent agents 3. To understand the concepts of artificial intelligence including problem-solving, knowledge representation, reasoning, decision making, planning, perception, and action. 4. To master the fundamentals of machine learning, mathematical framework, and learning algorithm. 5. Understand concepts of Spark, Zookeeper, and HBase					
Course Content	**List of Experiments:** 1. Study of Prolog. 2. Write simple facts for the statements using PROLOG. 3. Write predicates One converts centigrade temperatures to Fahrenheit, and the other checks if a temperature is below freezing. 4. Write a program to solve the Monkey Banana problem. 5. WAP in turbo prolog for medical diagnosis and show the advantages and disadvantages of green and red cuts. 6. WAP to implement factorial, fibonacci of a given number. 7. Write a program to solve the 4-Queen problem. 8. Write a program to solve the traveling salesman's problems. 9. Write a program to solve the water jug problem using LISP 10. Case Study and Project work.					
Textbooks	T1: Hadoop: The Definitive Guide, 4th Edition by Tom White - Shroff Publishers & Distributers Private Limited - Mumbai; Fourth edition (2015) T2: Basics of Artificial Intelligence & Machine Learning, Dr. Dheeraj Mehrotra					
References	R1: Big Data: Principles and Best Practices of Scalable Realtime Data Systems by James Warren and Nathan Marz, Manning Publications (2015) R2: Kenneth A. Lambert, "Fundamentals of Python – First Programs", CENGAGE Publication					
Course Outcomes	**After completing this course, students will be able to:** CO1. Understand the concepts of Hadoop and HDFS CO2. Understand the concepts of MapReduce CO3. Analyze big data tools Pig and Hive. CO4. Understand concepts of Spark, Zookeeper and HBase CO5. Perform load balancing and serialization in Hadoop.					

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3	Knowledge Levels (K_1, K_2, ..., K_6)
CO1	2	2	2								1	1	1	1	K_1, K_2, K_3
CO2	2	2	2	1							1	3	2	2	K_1, K_2, K_3, K_4
CO3	3	3	2	1	3						2	3	3	2	K_1, K_2, K_3, K_4
CO4	3	3	2	1	3						2	3	3	2	K_1, K_2, K_3, K_4
CO5	3	3	3	1	3						2	3	3	2	K_1, K_2, K_3, K_4, K_5

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

K_6 => Create

SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL						
School Name: School of Engineering & Technology			**Department Name: Advance Computing**			
Recommended Programs: MTech. in Artificial Intelligence					Semester: I	
Course Name	**Programming in PYTHON**		**Course Code**	**ET25MTAI115P**		
Credit Hours	**L**	**T**	**P**	**N**	**Total Credits**	2
	0	0	4	0		
Prerequisites	• Fundamentals of computing • Fundamentals of programming					
Course Objectives	In this course student will learn: 1. How to write simple programs using Python, a dynamic object-oriented programming language that can be used for many kinds of software development projects. 2. To apply data structures and programming constructs in the Python language. 3. To apply the object-oriented programming concepts. 4. To demonstrate the use of exception handling and file handling. 5. To apply the concept of multithreading and multiprocessing in Python.					
Course Content	**Program List** 1. Personal Information Write a program that displays the following information: • Your name • Your address, with the city, state, and ZIP • Your telephone number • Your college major 2. Loan Qualifier Program to determine whether a bank customer qualifies for a loan. 3. Day of the Week Write a program that asks the user for a number in the range of 1 through 7. The program should display the corresponding day of the week, where 1 = Monday, 2 = Tuesday, 3 = Wednesday, 4 = Thursday, 5 = Friday, 6 = Saturday, and 7 = Sunday. The program should display an error message if the user enters a number that is outside the range of 1 through 7. 4. Areas of Rectangles The area of a rectangle is the rectangle's length times its width. Write a program that asks for the length and width of two rectangles. The program should tell the user which rectangle has the greater area, or if the areas are the same. 5. Age Classifier Write a program that asks the user to enter a person's age. The program should display a message indicating whether the person is an infant, a child, a teenager, or an adult. Following are the guidelines: • If the person is 1 year old or less, he or she is an infant.					

	• If the person is older than 1 year but younger than 13 years, he or she is a child. • If the person is at least 13 years old, but less than 20 years old, he or she is a teenager. • If the person is at least 20 years old, he or she is an adult. 6. The program demonstrates creating a list, appending, inserting, removing, sorting, copying, and how to get the index of an item in a list and then replace that item with a new item. 7. Program to demonstrate the use of tuples 8. The program demonstrates creating a list, appending, inserting, removing, sorting, copying elements, access elements in the dictionary. 9. Program to demonstrate the use of strings 10. Sales Commission-Program calculates sales commissions. 11. The program demonstrates how the range function can be used with a for a loop. 12. Print asterisks in a pattern that looks like a triangle. 13. Program to navigate in different data structures 14. The program writes three lines of data to a file. 15. Program to create car and employee class 16. Program to create a class with initializer 17. Program to create an object of the class and use it 18. Program to inherit one class from parent 19. the program demonstrates polymorphism 20. Write a program to open a file and ask the user to input the filename. If the file exists, displays the content of the file, else create a file and write the line "File is opened in write mode". 21. program for the above exercise so it handles the following exceptions: • It should handle any IOError exceptions that are raised when the file is opened and data is read from it. 22. It should handle any ValueError exceptions that are raised when the items that are read from the file are converted to a number. 23. Write a statement that uses a math module function to convert 45 degrees to radians and assigns the value to a variable 24. write a program that calculates the following: • The area of a circle • The circumference of a circle • The area of a rectangle • The perimeter of a rectangle 25. The program demonstrates the sqrt function. 26. Program to import own packages 27. Ten Library functions of NumPy and pandas

Text Books	T1: Starting Out with Python, Toney Gaddis, Third Edition Pearson. T2: Python Data Science Handbook, Jake Vanderplas,
References	R1: Python Data Science Handbook, Jake Vanderplas, R2: Starting Out with Python, Toney Gaddis, Third Edition Pearson.
Course Outcomes	**After completion of this course, students are able to:** CO1Comprehend the basics of Python variables, data types, I/O functions, selection, and iteration. CO2.Apply data structures and programming constructs in the Python language. CO3.Apply the object-oriented programming concepts. CO4.Demonstrate the use of exception handling and file handling. CO5.Apply the concept of multithreading and multiprocessing in Python.

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PS O1	PS O2	PS O3	Knowledge Levels (K_1, K_2, ..., K_6)
CO 1	1	1	1	1	2						1	1	1	1	K_1, K_2
CO 2	2	1	1	1	1						1	2		1	K_1, K_2
CO 3	2	2	1	1	1					1	1	2	1		K_1, K_2, K_3
CO 4	3	2	2	1	2				1		2	2	1	2	K_1, K_2, K_3, K_4
CO 5	3	3	2	1	2			1		1	2	2	2	2	K_1, K_2, K_3, K_4

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

K_6 => Create

<table>
<tr><td colspan="9">SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL</td></tr>
<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="6">Department Name: Advance Computing</td></tr>
<tr><td colspan="7">Recommended Programs: M. Tech. in Artificial Intelligence</td><td colspan="2">Semester: I</td></tr>
<tr><td>Course Name</td><td colspan="2">Programming in Java</td><td>Course Code</td><td colspan="5">ET25MTAI116P</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2">Total Credits</td><td rowspan="2" colspan="3">2</td></tr>
<tr><td>0</td><td>0</td><td>4</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="8">● Fundamentals of computing
● Fundamentals of programming</td></tr>
<tr><td>Course Objectives</td><td colspan="8">Objective of this course is to:
1. Improve the analytical skills of object-oriented programming.
2. Overall development of problem-solving and critical analysis.
3. Formal introduction to Java programming language.
4. Demonstrate the concepts of polymorphism and inheritance.
5. Implement error handling techniques using exception handling.</td></tr>
<tr><td>Course Content</td><td colspan="8">List of experiments:
1. Program to define a structure of a basic JAVA program
2. Program to define the data types, variables, operators, arrays, and control structures.
3. Program to define class and constructors. Demonstrate constructors.
4. Program to define class, methods, and objects. Demonstrate method overloading.
5. Program to define inheritance and show method overriding.
6. Program to demonstrate Packages.
7. Program to demonstrate Exception Handling.
8. Program to demonstrate Multithreading.
9. Program to demonstrate I/O operations
10. Program to demonstrate Applet structure and event handling</td></tr>
<tr><td>Text Books</td><td colspan="8">T1: Timothy Budd, “An Introduction to Object-Oriented Programming”, Addison-Wesley Publication, 3rd Edition
T2: Cay S. Horstmann and Gary Cornell, “Core Java: Volume I, Fundamentals”, Prentice-Hall publication.</td></tr>
<tr><td>References</td><td colspan="8">R1: G. Booch, “Object-Oriented Analysis& Design”, Addison Wesley.
R2: James Martin, “Principles of Object-Oriented Analysis and Design”, Prentice-Hall/PTR
R3: Peter Coad and Edward Yourdon, “Object-Oriented Design”, Prentice-Hall/PTR.</td></tr>
<tr><td>Course Outcomes</td><td colspan="8">After completion of this course, students are able to:
CO 1. Comprehend the basics of Java programming language.
CO 2. Identify classes, objects, members of a class and relationships.
CO 3. Write Java application programs using OOP principles.
CO 4. Demonstrate the concepts of polymorphism and inheritance.
CO 5.Implement error handling techniques using exception handling.</td></tr>
</table>

Mapping of Course outcome with Program Outcomes, PSO’s, and Knowledge Levels (As per Blooms Taxonomy)

CO/ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, …, K_6)
CO1	2	2	2								2	1	1	1	**K_1, K_2, K_3**
CO2	2	2	2	1							2	3	2	2	**K_1, K_2, K_3, K_4**
CO3	3	3	2	1	3						2	3	3	2	**K_1, K_2, K_3, K_4**
CO4	3	3	2	1	3						2	3	3	2	**K_1, K_2, K_3, K_4**
CO5	3	3	3	1	3						2	3	3	2	**K_1, K_2, K_3, K_4, K_5**

High-3 Medium-2 Low-1

K_1 =>Remember K_2 =>Understand K_3 =>Apply K_4 =>Analyze K_5 =>Evaluate

K_6 =>Create

<table>
<tr><td colspan="8">SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL</td></tr>
<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="5">Department Name: Advance Computing</td></tr>
<tr><td colspan="5">Recommended Program: MTech. in Artificial Intelligence</td><td colspan="3">Semester: I</td></tr>
<tr><td>Course Name</td><td colspan="2">LISP Programming</td><td>Course Code</td><td colspan="4">ET25MTAI117P</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2">Total Credits</td><td rowspan="2" colspan="2">2</td></tr>
<tr><td>0</td><td>0</td><td>4</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="7">• Fundamentals of computing
• Fundamentals of programming</td></tr>
<tr><td>Course Objectives</td><td colspan="7">1. Improve the analytical skills of object-oriented programming.
2. Implement an expert system using LISP.
3. Implement NLP using Lex
4. Overall development of problem-solving and critical analysis.
5. Formal introduction to Java programming language.</td></tr>
<tr><td>Course Content</td><td colspan="7">List of experiments:
1. Implement these practicals in LISP in which you feel comfortable.
a) Depth –bounded depth-first search.
b) Iterative Deepening Search.
c) Best first search.
d) A * Search.
e) AO* Search.
f) Minmax Search.
g) Alpha Beta Pruning.
2. Solve the water jug problem using the AI technique.
3. Solve the Missionaries problem using AI technique.
4. Design the following expert system using LISP in which you feel comfortable.
a) Weather Forecasting System.
b) Legal Expert System.
5. Design parser for NLP using Lex and Yacc utilities</td></tr>
<tr><td>Text Books</td><td colspan="7">T1: David S. Touretzky, “Common LISP: A Gentle Introduction to Symbolic Computation”
T2: Paul Graham, “On Lisp Advance Technology for common Lisp”,</td></tr>
<tr><td>References</td><td colspan="7">R1: Christian Queinnec, “Lisp in Small Pieces.”
R2: Peter Seibel, “Practical Common Lisp”, Prentice Hall/PTR</td></tr>
<tr><td>Course Outcomes</td><td colspan="7">After completion of this course, students are able to:
CO 1. Comprehend the basics of the Lisp programming language.
CO 2. The student will be able to handle applications in Tree and Graph.
CO 3. Comprehend the basics of AI technique.
CO 4. Implement an expert system using LISP.
CO 5. Implement NLP using Lex</td></tr>
</table>

Mapping of Course outcome with Program Outcomes, PSO’s, and Knowledge Levels (As per Blooms Taxonomy)

CO/ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, …, K_6)
CO 1	1	1	1	1	2						1	1	1	1	**K_1, K_2**
CO 2	2	1	1	1	1						1	2		1	**K_1, K_2**
CO 3	2	2	1	1	1					1	1	2	1		**K_1, K_2, K_3**
CO 4	3	2	2	1	2				1		2	2	1	2	**K_1, K_2, K_3, K_4**
CO 5	3	3	2	1	2			1		1	2	2	2	2	**K_1, K_2, K_3, K_4**

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

K_6 => Create

SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL

Proposed Scheme & Syllabus

for

Master of Technology (Artificial Intelligence)

II Semester

School of Engineering and Technology

<table>
<tr><td colspan="8">SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL</td></tr>
<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="5">Department Name: Advance Computing</td></tr>
<tr><td colspan="5">Recommended Programs: M. Tech. in Artificial Intelligence</td><td colspan="3">Semester: II</td></tr>
<tr><td>Course Name</td><td colspan="2">Computational Perception</td><td>Course Code</td><td colspan="4">ET25MTAI201T</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2" colspan="2">Total Credits</td><td rowspan="2">4</td></tr>
<tr><td>4</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="7">1.Artificial Intelligence & Foundation of ML
2.Mathematics for Computing
3.Computability, Algorithms, and Complexity</td></tr>
<tr><td>Course Objectives</td><td colspan="7">1.To learn advanced aspects of perception and scene analysis in both the visual and auditory modalities.
2.Learn the experimental approaches of scientific disciplines and the computational approaches of engineering disciplines.
3.To understand the set of fundamental computational problems that must be solved in robust perceptual systems.
4.To learn the sensory coding, perceptual invariance, spatial vision, sound localization, and visual and auditory scene segmentation.
5.To study many aspects of attention, and the basics of recognition in natural visual and auditory scenes.</td></tr>
<tr><td rowspan="5">Course Content</td><td colspan="6">Unit 1:
Sound localization, linear systems theory, Bayesian Inference, Auditory sensory coding, information, Visual sensory coding, and information theory.</td><td>12 Hours</td></tr>
<tr><td colspan="6">Unit 2:
Computation and representation of visual motion, Perceptual inference, Bayesian modeling, Visual structure, representation of shape and surfaces</td><td>12 Hours</td></tr>
<tr><td colspan="6">Unit 3:
Perceptual constancy, Auditory structure, Auditory scene analysis, Eye movements, Visual search, Visual scene analysis, Perceptual organization, Object recognition, and class retrospective</td><td>12 Hours</td></tr>
<tr><td colspan="6">Unit 4:
Perception (uni- and multi-modal) Vision, Hearing, taste, touch, olfaction, Higher-order perception like object and pattern recognition (Biological and computational perspective) Visual Perception of 3D space & scene Perceptual processes for object recognition & memorization</td><td>12 Hours</td></tr>
<tr><td colspan="6">Unit 5:
Attentional mechanism in multimedia perception, modeling perception, cognition & behavior in real-time applications, Executive function like planning & monitoring of complex behavior, learning, memory and knowledge representations, Real-world Applications (In Technology development, Health care, Defense, Industries, etc.)</td><td>12 Hours</td></tr>
<tr><td>Text Books</td><td colspan="7">T1: Nakayama, K. (1998). Vision fin-de-Siecle - a reductionistic explanation of perception for the 21st century? In Hochberg, J., editor, Perception, and Cognition at Century’s End: History, Philosophy, Theory, pages 307–331. Academic Press.
T2: Von B´ek´esy, G. (1960). Experiments in Hearing, chapter 1: The Problems of Auditory Research, pages 3–10. McGraw-Hill.</td></tr>
</table>

References	R1: Barlow, H. B. (1994). What is the computational goal of neocortex? In Koch, C. and Davis, J. L., editors, Large-scale neuronal theories of the brain, pages 1–22. MIT Press. R2: Churchland, P.S., Ramachandran,V.S.,and Sejnowski,T.J.(1994). A critique of pure vision. In Koch,C.and Davis,J.L.,editors, Large-scale neuronal theories of the brain, pages 23–60. MIT Press. R3: Marr,D.(1982). Vision, chapter1: The Phiosophy and the Approach, pages8–38. Freeman. Richards,W.(1988). The approach. InRichards,W.,editor, Natural Computation, pages 3–13. MIT Press.
Course Outcomes	**After completing this course, students will be able to:** CO1. omprehend the advanced aspects of perception and scene analysis in both the visual and auditory modalities, concentrating on those aspects that allow us and animals to behave in natural, complex environments. CO2. Able to identify experimental approaches of scientific disciplines and the computational approaches of engineering disciplines. CO3. Understand set of fundamental computational problems that must be solved in robust perceptual systems. CO4. Scientific reasoning and key experimental results that lead to our current understanding of the important computational problems in perception and scene analysis. CO5. Understanding of the sensory coding, perceptual invariance, spatial vision and sound localization, visual and auditory scene segmentation.

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, …, K_6)
CO 1	1	1	1	1	2					1	1	1	1	1	K_1, K_2
CO 2	2	1	1	1	1						1	2		1	K_1, K_2
CO 3	2	2	1	1	1					1	1	2	1		K_1, K_2, K_3
CO 4	3	2	2	1	2				1		2	2	1	2	K_1, K_2, K_3, K_4
CO 5	3	3	2	1	2			1		1	2	2	2	2	K_1, K_2, K_3, K_4

High-3 **Medium-2** **Low-1**

K_1 => Remember **K_2 => Understand** **K_3 => Apply** **K_4 => Analyze** **K_5 => Evaluate**

K_6 => Create

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<tr><td colspan="4">School Name: School of Engineering & Technology</td><td colspan="4">Department Name: Advance Computing</td></tr>
<tr><td colspan="6">Recommended Programs: M. Tech. in Artificial Intelligence</td><td colspan="2">Semester: II</td></tr>
<tr><td>Course Name</td><td colspan="2">Deep learning</td><td>Course Code</td><td colspan="4">ET25MTAI202T</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2" colspan="2">Total Credits</td><td rowspan="2">3</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="7">1.Artificial Intelligence & Foundation of ML
2.Programming in PYTHON
3.Artificial Neural Networks</td></tr>
<tr><td>Course Objectives</td><td colspan="7">1.Become an expert in neural networks
2.Learn to implement NNs in Keras and TensorFlow.
3.Build convolutional networks for image recognition.
4.Build recurrent networks for sequence generation.
5.Build generative adversarial networks for image generation.</td></tr>
<tr><td rowspan="5">Course Content</td><td colspan="6">Unit- I
Introduction To Neural Networks, Implementing Gradient Descents Training Neural Networks, Sentiment Analysis, Keras, TensorFlow</td><td>10 Hours</td></tr>
<tr><td colspan="6">Unit- II
Convolutional Neural Network, CNNs in TensorFlow, Weight Initialization, Autoencoders, Transfer Learning in TensorFlow, Deep Learning for Cancer Detection</td><td>10 Hours</td></tr>
<tr><td colspan="6">Unit- III
Recurrent Neural Networks, Long Short-Term Memory Network Implementation Of RNN And LSTM, Hyperparameters, Embeddings and Word2vec, Sentiment Prediction RNN</td><td>09 Hours</td></tr>
<tr><td colspan="6">Unit- IV
Generative Adversarial Network, Deep Convolutional GANs, Generate Faces, Semi-Supervised Learning, The RL Framework: The Problem, The RL Framework: The Solution, Dynamic Programming</td><td>08 Hours</td></tr>
<tr><td colspan="6">Unit- V
Monte Carlo Methods, Temporal-Difference Methods, RL In Continuous Spaces, Deep Q-Learning, Policy Gradients, Actor-Critic Methods</td><td>08 Hours</td></tr>
<tr><td>Text Books</td><td colspan="7">T1: Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 2nd Edition, Pearson Education.
T2: Elaine Rich, Kevin Knight, Shivshankar B Nair, Artificial Intelligence, McGraw Hill, 3rd Edition.
T3: Elaine Rich, Kevin Knight, Artificial Intelligence, Tata McGraw Hill, 2nd Edition.</td></tr>
<tr><td>References</td><td colspan="7">R1: Lugar, .AI-Structures and Strategies for Complex Problem Solving., 4/e, 2002, Pearson Education.
R2: Nils J. Nilsson, Principles of Artificial Intelligence, Narosa Publication.
R3: Patrick H. Winston, Artificial Intelligence, 3rd edition, Pearson Education.
R4: Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Publication</td></tr>
<tr><td>Course Outcomes</td><td colspan="7">After completing this course, students will be able to:
CO1.Students will develop a basic understanding of the building blocks of AI as presented in terms of intelligent agents.
CO2.Students will be able to choose an appropriate problem-solving method and knowledge-representation scheme.</td></tr>
</table>

	CO3.Students will develop an ability to analyze and formalize the problem (as a state space, graph, etc.) and select the appropriate search method. CO4.Students will be able to develop/demonstrate/ build simple intelligent systems CO5.Students will be able to develop/demonstrate/ build various classical toy problems using different AI techniques.

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, …, K_6)
CO 1	1	1	1	1	2						1	1	1	1	**K_1, K_2**
CO 2	2	1	1	1	1					1	1	2		1	**K_1, K_2**
CO 3	2	2	1	1	1						1	2	1		**K_1, K_2, K_3**
CO 4	3	2	2	1	2				1		2	2	1	2	**K_1, K_2, K_3, K_4**
CO 5	3	3	2	1	2			1		1	2	2	2	2	**K_1, K_2, K_3, K_4**

High-3 **Medium-2** **Low-1**

K_1 => Remember **K_2 => Understand** **K_3 => Apply** **K_4 => Analyze** **K_5 => Evaluate**

K_6 => Create

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<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="5">Department Name: Advance Computing</td></tr>
<tr><td colspan="5">Recommended Programs: M.Tech. in Artificial Intelligence</td><td colspan="3">Semester: II</td></tr>
<tr><td>Course Name</td><td colspan="2">Artificial Intelligence for Robotics</td><td>Course Code</td><td colspan="4">ET25MTAI211T/P</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2">Total Credits</td><td rowspan="2" colspan="2">3</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="7">1.Artificial Intelligence & Foundation of ML
2.Programming in PYTHON
3.Artificial Neural Networks</td></tr>
<tr><td>Course Objectives</td><td colspan="7">1.Leverage fundamentals of Al and robotics
2.Work through use cases to implement various machine learning algorithms
3.Explore Natural Language Processing (NLP) concepts for efficient decision-making in robots
4.Design your robot to listen using NLP via an expert system
5.Use machine learning and computer vision to teach your robot how to avoid obstacles</td></tr>
<tr><td rowspan="5">Course Content</td><td colspan="6">Unit-I:
Introduction to Robotics, types of Robots, Robot Anatomy-Definition, law of robotics, History and Terminology of Robotics-Accuracy and repeatability of Robotics-Simple problems Specifications of Robot, Robot classifications Architecture of robotic systems</td><td>10 Hours</td></tr>
<tr><td colspan="6">Unit-II
Foundation for Advanced Robotics and AI, Basic principles of robotics and AI, Setting Up Your Robot-technical requirements, Software Setup, Hardware, , A Concept for a Practical Robot Design Process, Object Recognition Using Neural Networks and Supervised Learning</td><td>10 Hours</td></tr>
<tr><td colspan="6">Unit-III
Picking up the Toys, Technical Requirement, Teaching the robot arm, Pick up objects using genetic algorithms
Course Content</td><td>09 Hours</td></tr>
<tr><td colspan="6">Unit-IV
Robot to Listen-Technical specifications, Robot speech recognition, Avoiding the Stairs- Technical specifications, Task Analysis</td><td>08 Hours</td></tr>
<tr><td colspan="6">Unit- V
Putting Things Away, Giving the Robot an Artificial Personality</td><td>08 Hours</td></tr>
<tr><td>Text Books</td><td colspan="7">T1: Artificial Intelligence for Robotics: Build intelligent robots that perform human tasks using AI techniques, 2018 by Francis X. Govers, Packt
T2: Robotics, Vision and Control: Fundamental Algorithms In MATLAB, Second Edition (Springer Tracts in Advanced Robotics) 2nd ed. 2017 Edition by Peter Corke, Springer</td></tr>
<tr><td>References</td><td colspan="7">R1: https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-to robotics-fall-2005/lecture-notes/
R2: Robotics, Vision and Control: Fundamental Algorithms In MATLAB, Second Edition (Springer Tracts in Advanced Robotics) 2nd ed. 2017 Edition by Peter Corke, Springer</td></tr>
<tr><td>Course Outcomes</td><td colspan="7">After completing this course, students will be able to:
CO 1. Understand path planning, decision trees, and search algorithms in order to enhance your robot, and apply simulation techniques to give your robot an artificial personality
CO 2. Understand object recognition using neural networks and supervised learning</td></tr>
</table>

	techniques CO 3. Pick up objects using genetic algorithms for manipulation CO 4. Teach your robot to listen using NLP via an expert system CO 5. Use machine learning and computer vision to teach your robot how to avoid obstacles

Mapping of Course outcomes with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, …, K_6)
CO 1	1	1	1	1	2						1	1	1	1	K_1, K_2
CO 2	2	1	1	1	1						1	2		1	K_1, K_2
CO 3	2	2	1	1	1						1	2	1		K_1, K_2, K_3
CO 4	3	2	2	1	2				1		2	2	1	2	K_1, K_2, K_3, K_4
CO 5	3	3	2	1	2			1		1	2	2	2	2	K_1, K_2, K_3, K_4

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

K_6 => Create

<table>
<tr><th colspan="8">SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL</th></tr>
<tr><td colspan="4">School Name: School of Engineering & Technology</td><td colspan="4">Department Name: Advance Computing</td></tr>
<tr><td colspan="6">Recommended Programs: M.Tech. in Artificial Intelligence</td><td colspan="2">Semester: II</td></tr>
<tr><td>Course Name</td><td colspan="2">Digital Image Processing</td><td>Course Code</td><td colspan="4">ET25MTAI212T/P</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2" colspan="2">Total Credits</td><td rowspan="2">3</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="7">1.Mathematics for Computing
2.Artificial Intelligence & Foundation of ML
3.Programming in PYTHON</td></tr>
<tr><td>Course Objectives</td><td colspan="7">1.To study the fundamentals of digital image processing
2.To study the image transforms and enhancement techniques.
3.To study image restoration and construction techniques.
4.To study image compression and segmentation
5.To study color and multispectral image processing.</td></tr>
<tr><td rowspan="5">Course Content</td><td colspan="6">Unit – I
Digital image fundamentals Introduction: Digital Image- Steps of Digital Image Processing Systems-Elements of Visual Perception - Connectivity and Relations between Pixels. Simple Operations- Arithmetic, Logical, Geometric Operations. Mathematical Preliminaries - 2D Linear Space Invariant Systems - 2D Convolution - Correlation 2D Random Sequence - 2D Spectrum.</td><td>10 Hours</td></tr>
<tr><td colspan="6">Unit – II
Image transforms and enhancement Image Transforms: 2D Orthogonal and Unitary Transforms-Properties and Examples. 2D DFT- FFT – DCT - Hadamard Transform - Haar Transform - Slant Transform - KL Transform - Properties And Examples.Image Enhancement: - Histogram Equalization Technique- Point Processing-Spatial Filtering-In Space And Frequency - Nonlinear Filtering-Use Of Different Masks.</td><td>10 Hours</td></tr>
<tr><td colspan="6">Module – III
Image restoration and construction Image Restoration: Image Observation and Degradation Model, Circulant And Block Circulant Matrices and Its Application In Degradation Model - Algebraic Approach to Restoration-Inverse By Wiener Filtering - Generalized Inverse-SVD And Interactive Methods - Blind Deconvolution-Image Reconstruction From Projections.</td><td>09 Hours</td></tr>
<tr><td colspan="6">Module – IV
Image compression & segmentation Image Compression: Redundancy and Compression Models -Loss Less and Lossy. Loss Less- Variable-Length, Huffman, Arithmetic Coding - Bit-Plane Coding, Loss Less Predictive Coding, Lossy Transform (DCT) Based Coding, JPEG Standard - Sub Band Coding. Image Segmentation: Edge Detection - Line Detection - Curve Detection - Edge Linking and Boundary Extraction, Boundary Representation, Region Representation and Segmentation</td><td>08 Hours</td></tr>
<tr><td colspan="6">Module – V
Color and multispectral image processing Color Image-Processing Fundamentals, RGB Models, HSI Models, Relationship Between Different Models. Multispectral Image Analysis - Color Image Processing Three-Dimensional Image Processing-Computerized Axial Tomography-Stereometry-Stereoscopic Image Display-Shaded Surface Display</td><td>08 Hours</td></tr>
</table>

Text Books	T1: Digital Image Processing, Gonzalez.R.C & Woods. R.E., 3/e, Pearson Education, 2008. T2: Digital Image Processing, Kenneth R Castleman, Pearson Education,1995. T3: Digital Image Procesing, S. Jayaraman, S. Esakkirajan, T. Veerakumar, McGraw Hill Education ,2009. Pvt Ltd, NewDelhi
References	R1: mentals of Digital image Processing, Anil Jain.K, Prentice Hall of India, 1989. R2: Image Processing, Sid Ahmed, McGraw Hill, New York, 1995.
Course Outcomes	**After completing this course, students will be able to:** CO1.Students will develop a basic understanding and connectivity of Digital Image Processing Systems CO2.Students will be able to different Image transforms system having different frequency images CO3.Students will develop Image restoration and construction Image Restoration by using several methods. CO4.Students will be able to develop/demonstrate/ build Image compression & segmentation of Image Compression CO5.Students will be able to develop/demonstrate/ build various Color Image-Processing using different techniques.

Mapping of Course outcomes with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, …, K_6)
CO1	1	1	1	1	2					1	1	1	1	1	K_1, K_2
CO2	2	1	1	1	1						1	2		1	K_1, K_2
CO3	2	2	1	1	1					1	1	2	1		K_1, K_2, K_3
CO4	3	2	2	1	2				1		1	2	1	2	K_1, K_2, K_3, K_4
CO5	3	3	2	1	2			1		1	1	2	2	2	K_1, K_2, K_3, K_4

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

K_6 => Create

SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL						
School Name: School of Engineering & Technology			**Department Name: Advance Computing**			
Recommended Programs: MTech. In Artificial Intelligence					Semester: II	
Course Name	**Deductive Systems**		**Course Code**	**ET25MTAI213T**		
Credit Hours	**L**	**T**	**P**	**N**	**Total Credits**	4
	4	0	0	0		
Prerequisites	1.Intro. To Problem Solving & Reasoning 2.Mathematics for Computing					
Course Objectives	1.To learn the principles of deductive and inductive reasoning. 2.To learn the formal language concepts 3.To study the fallacies and their applications. 4.To study the concepts of Heuristics 5.To study inductive inference methods.					
Course Content	**UNIT I – FORMAL LANGUAGE CONCEPTS** Some definitions of formal logical concepts- Classical symbolic logic – a symbolic representation of language statements – formal logical rules of inference – semantics in formal logic – provability relation – does formal logic model				**12 Hours**	
	UNIT II – HUMAN REASONING Human reasoning – mental model theory – revised model theory of conditionals - nonmonotonic logic - categorization and default reasoning – minimal model semantics-default entailment relation – some characteristic of belief - biases in human reasoning.				**12 Hours**	
	UNIT III – HEURISTICS The representativeness heuristics and the availability heuristics – Atmosphere effect – effects of negation - Introduction – Affirming the consequent and denying the antecedent – errors in the interpretation of standard form categorical propositions – fallacies due to ambiguity of language - language nuances associated with conditional statements – conditional inferences made of 'only if-statements - ordinary languages Vs formal language definitions of quantifiers.				**12 Hours**	
	UNIT IV – INDUCTIVE INFERENCE Nature of Inductive inference – method of agreement – method of difference – method of residues – method of concomitant variations – the argument from analogy – Imperfect applications of Inductive methods – relation of induction to deduction and verification.				**12 Hours**	
	UNIT V - FALLACIES OF INDUCTIVE REASONING Fallacies of generalization – Fallacies of non-observation – False analogy – interpreting asymmetries of projection in Children's inductive reasoning - use of single or multiple categories in category-based induction – abductive inference from philosophical analysis to neural mechanisms.				**12 Hours**	
Text Books	T1: Thomas Fowler "Logic: Deductive and Inductive", Adamant Media Corporation 2004 T2: Aidan Feeney and Evan Heit, "Inductive Reasoning: Experimental, developmental and computational approaches", Cambridge University Press, 2007.					
References	R1: Thomas Fowler "Logic: Deductive and Inductive", Adamant Media Corporation 2004. R2: Aidan Feeney and Evan Heit, "Inductive Reasoning: Experimental, developmental and computational approaches", Cambridge University Press, 2007.					

Course Outcomes	**After completing this course, students will be able to:** CO1.Understand the definitions and approaches to deductive reasoning. CO2.Comprehend the inductive methods and fallacies. CO3.Learn the fallacies and their applications. CO4.Understand the concepts of Heuristics CO5.Learn the inductive inference methods.

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, …, K_6)
CO 1	1	1	1	1	2						1	1	1	1	K_1, K_2
CO 2	2	1	1	1	1						1	2		1	K_1, K_2
CO 3	2	2	1	1	1					1	1	2	1		K_1, K_2, K_3
CO 4	3	2	2	1	2				1		2	2	1	2	K_1, K_2, K_3, K_4
CO 5	3	3	2	1	2			1		1	2	2	2	2	K_1, K_2, K_3, K_4

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

K_6 => Create

<table>
<tr><td colspan="8">SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL</td></tr>
<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="5">Department Name: Advance Computing</td></tr>
<tr><td colspan="6">Recommended Programs: MTech. In Artificial Intelligence</td><td colspan="2">Semester: II</td></tr>
<tr><td>Course Name</td><td colspan="2">Decision Making Under Uncertainty</td><td>Course Code</td><td colspan="4">ET25MTAI214T</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2" colspan="2">Total Credits</td><td rowspan="2">4</td></tr>
<tr><td>4</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="7">● Mathematics for Computing
● Computability, Algorithms, and Complexity
● Introduction to Problem Solving & Reasoning</td></tr>
<tr><td>Course Objectives</td><td colspan="7">1. Examine some of the techniques for modeling decision problems of various types
2. Study computational methods used to solve various decision problems.
3. To study the probabilistic models: probabilistic inference, decision making under uncertainty.
4. To study the game-theoretic models of multiagent interaction.
5. To learn Multiagent decision-making</td></tr>
<tr><td rowspan="5">Course Content</td><td colspan="6">Unit 1:
Probabilistic semantics for belief, Representation of probabilities, Bayesian networks, Inference in Bayesian networks, Preferences and utilities, Rational decision-making, Foundations of utility theory, multi-attribute utility theory.</td><td>12 Hours</td></tr>
<tr><td colspan="6">Unit 2:
Preference elicitation, Optimal planning, and multi-stage decision making, Markov decision processes, Structured computational techniques for MDPs., Function approximation and simulation, Partially observable Markov decision processes</td><td>12 Hours</td></tr>
<tr><td colspan="6">Unit 3:
Intro to Reinforcement Learning (time-permitting), Multiagent decision making: Game theory, Basics of game theory, Refinements of Nash equilibria, Stochastic and Markov games, Cooperation.</td><td>12 Hours</td></tr>
<tr><td colspan="6">Unit 4:
Games of incomplete information (Bayesian games), Mechanism design (and computational approaches to MD), Auction theory (as an illustration of mechanism design), Multiagent decision making: Social choice</td><td>12 Hours</td></tr>
<tr><td colspan="6">Unit 5:
Elements of social choice and voting, Voting rules, Computational considerations, Manipulation, and control, Voting with partial information, Matching problems, Other forms of "mechanism design without money".</td><td>12 Hours</td></tr>
<tr><td>Text Books</td><td colspan="7">T1: Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations by Yoav Shoham and Kevin Leyton-Brown (Cambridge University Press, 2009).
T2: rwiche. Bayesian Networks. Handbook of Knowledge Representation. Volume 3, 2008, Pages 467-509, Elsevier. Read from 11.1 to 11.3.2</td></tr>
<tr><td>References</td><td colspan="7">R1: R. Dechter, Bucket Elimination: A Unifying Framework for Probabilistic Inference. UAI-96. Read up to 3.3 (5 pages).</td></tr>
</table>

	R2: Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations by Yoav Shoham and Kevin Leyton-Brown (Cambridge University Press, 2009).
Course Outcomes	**After completing this course, students will be able to:** CO1.Examine some of the techniques for modeling decision problems of various types CO2.Understand computational methods used to solve various decision problems. CO3.Understand the probabilistic models: probabilistic inference, decision making under uncertainty. CO4.Understand the game-theoretic models of multiagent interaction. CO5.Learn Multiagent decision making

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PS O1	PS O2	PS O3	Knowledge Levels (K_1, K_2, ..., K_6)
CO 1	1	1	1	1	2						1	1	1	1	K_1, K_2
CO 2	2	1	1	1	1						1	2		1	K_1, K_2
CO 3	2	2	1	1	1						1	2	1		K_1, K_2, K_3
CO 4	3	2	2	1	2				1		2	2	1	2	K_1, K_2, K_3, K_4
CO 5	3	3	2	1	2			1		1	2	2	2	2	K_1, K_2, K_3, K_4

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

K_6 => Create

SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL							
School Name: School of Engineering & Technology			**Department Name: Advance Computing**				
Recommended Programs: M. Tech. in Artificial Intelligence					Semester: II		
Course Name	**Data Privacy & User Protection**		**Course Code**	**ET25MTAI215T**			
Credit Hours	**L**	**T**	**P**	**N**	**Total Credits**	4	
	4	0	0	0			
Prerequisites	1.Fundamentals of cyber security 2.Fundamentals of information security						
Course Objectives	1.To study security and privacy with a set of techniques. 2.To address the security & privacy challenges. 3.Demonstrate various measurement, privacy, and enforcement policies 4.Use privacy tools during internet surfing 5.Implement privacy tools web service						
Course Content	**Unit I: Introduction Data Privacy:** Introduction Data Privacy, The increasing vulnerabilities of networks, Privacy issues, Privacy cost assessments in cyber-attack, privacy-enhancing technologies, VPN, Privacy in RFID, Privacy in cyber-physical systems and Internet of Things (IoT).					**12 Hours**	
	Unit II: Privacy policies and Enforcement: Privacy measurement, Privacy policies and Enforcement, P3PPolicy, Policy Enforcement Techniques, Issues, Models of Privacy, Machine Readable Policy.					**12 Hours**	
	Unit III: Internet Privacy: Internet Privacy, Importance of good password, minimizing your Online Digital Fingerprints, Preventing Identity Theft and Fraud, Risk.					**12 Hours**	
	Unit IV: Web Privacy: Dangers of Wireless Networks and "Hotspots", Securing Devices, Using Encryption to Hide and Keep Safe your Personal Digital Items, Information, and Data, Torrent File Sharing and Anonymous on the web, Secure, Private and Anonymous Usenet.					**12 Hours**	
	Unit V: Social Networks Privacy and security: Social Networks Privacy and security on social media. Threats to privacy in OSNs Threats regarding awareness, control, trustworthiness. Children's Privacy, Online Advertising, Digital Afterlife, Health, and Genetic Privacy.					**12 Hours**	
Text Books	T1: Complete guide to Internet Privacy, Anonymity and Security (Matthew Bailey). T2: Internet Privacy: Options for adequate realization edited (Johannes Buchman).						
References	R1: Complete guide to Internet Privacy, Anonymity and Security (Matthew Bailey). R2: Internet Privacy: Options for adequate realization edited (Johannes Buchman).						
Course Outcomes	**After completing this course, students will be able to:** CO1: Understand the concept of data privacy CO2: Demonstrate various measurement, privacy, and enforcement policies CO3: Use privacy tools during internet surfing CO4: Implement privacy tools web service CO5: Analyze privacy in Social Networks						

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PSO 1	PSO 2	PSO 3	Knowledge Levels ($K_1, K_2, \ldots, K_6$)
CO1	1	1	1	1	2						1	1	1	1	**K_1, K_2**
CO2	2	1	1	1	1					1	1	2		1	**K_1, K_2**
CO3	2	2	1	1	1					1	1	2	1		**K_1, K_2, K_3**
CO4	3	2	2	1	2				1		2	2	1	2	**K_1, K_2, K_3, K_4**
CO5	3	3	2	1	2			1		1	2	2	2	2	**K_1, K_2, K_3, K_4**

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

K_6 => Create

SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL

School Name: School of Engineering & Technology	**Department Name: Advance Computing**
Recommended Programs: M. Tech. in Artificial Intelligence	Semester: II

Course Name	**Information Retrieval**	**Course Code**	**ET25MTAI216T**

Credit Hours	**L**	**T**	**P**	**N**	**Total Credits**
	4	0	0	0	4

Prerequisites	PIG Programming	
Course Objectives	1. To study the various information retrieval models. 2. To study the web search engine and perform Link analysis. 3. To understand Hadoop and Map Reduce. 4. To learn document text mining techniques	
Course Content	**UNIT I: INTRODUCTION** Introduction -History of IR- Components of IR - Issues –Open source Search engine Frameworks - The impact of the web on IR - The role of artificial intelligence (AI) in IR – IR Versus Web Search - Components of a Search engine-Characterizing the web.	**10 Hours**
	UNIT II: INFORMATION RETRIEVAL Boolean and vector-space retrieval models- Term weighting - TF-IDF weighting- cosine similarity – Preprocessing - Inverted indices - efficient processing with sparse vectors – Language Model-based IR - Probabilistic IR –Latent Semantic Indexing - Relevance feedback and query expansion.	**08 Hours**
	UNIT III: WEB SEARCH ENGINE – INTRODUCTION AND CRAWLING Web search overview, web structure, the user, paid placement, search engine optimization/ spam. Web size measurement - search engine optimization/spam – Web Search Architectures - crawling - meta-crawlers- Focused Crawling - web indexes –- Near-duplicate detection - Index Compression - XML retrieval.	**10 Hours**
	UNIT IV: WEB SEARCH – LINK ANALYSIS AND SPECIALIZED SEARCH Link Analysis –hubs and authorities – Page Rank and HITS algorithms - Searching and Ranking – Relevance Scoring and ranking for Web – Similarity - Hadoop & Map Reduce - Evaluation - Personalized search - Collaborative filtering and content-based recommendation of documents and products – handling "invisible" Web - Snippet generation, Summarization, Question Answering, Cross-Lingual Retrieval.	**08 Hours**
	UNIT V: DOCUMENT TEXT MINING Information filtering; organization and relevance feedback – Text Mining - Text classification and clustering - Categorization algorithms: naive Bayes; decision trees; and nearest neighbor - Clustering algorithms: agglomerative clustering; k-means; expectation maximization (EM)	**09 Hours**
Text Books	T1. C. Manning, P. Raghavan, and H. Schütze, Introduction to Information Retrieval, Cambridge University Press, 2008. T2. Ricardo Baeza -Yates and Berthier Ribeiro - Neto, Modern Information Retrieval: The Concepts and Technology behind Search 2 nd Edition, ACM Press Books 2011. T3. Bruce Croft, Donald Metzler, and Trevor Strohman, Search Engines: Information Retrieval in Practice, 1 st Edition Addison Wesley, 2009. T4. Mark Levene, An Introduction to Serach Engines and Web Navigation, 2nd Edition Wiley, 2010	
References	R1: Stefan Boettcher, Charles L. A. Clarke, Gordon V. Cormack, Information Retrieval: Implementing and Evaluating Search Engines, The MIT Press, 2010. R2: Ophir Frieder "Information Retrieval: Algorithms and Heuristics: The Information Retrieval	

	Series ", 2 nd Edition, Springer, 2004 R3: Manu Konchady, "Building Search Applications: Lucene, Ling Pipe", and First Edition, Gate Mustru Publishing, 2008.
Course Outcomes	**After completing this course, students will be able to:** **CO1**: To understand the components of Information retrieval and IR vs web search. **CO2:** To understand the statistical models of text that can be used for IR applications **CO3:** To understand **w**eb search engines and crawling. **CO4:** To understand the link analysis and specialized search. **CO5:** To Apply and analyze IR techniques for text classification and clustering.

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PSO 1	PSO 2	PSO 3	Knowledge Levels (K_1, K_2, ..., K_6)
CO1		2	2	1	3								2		2
CO2		2	3	1	2								2		2
CO3		2	2	2	3								2		3
CO4		1	2	2	3								2	3	2
CO5		3	2	2	3							2	2	3	3

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

K_6 => Create

<table>
<tr><td colspan="8">SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL</td></tr>
<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="5">Department Name: Advance Computing</td></tr>
<tr><td colspan="6">Recommended Programs: MTech. In Artificial Intelligence</td><td colspan="2">Semester: II</td></tr>
<tr><td>Course Name</td><td colspan="2">Deep learning Lab</td><td>Course Code</td><td colspan="4">ET25MTAI202P</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2">Total Credits</td><td colspan="2" rowspan="2">1</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="7">1.Artificial Intelligence & Foundation of ML (ACTDCAIM003T/P)
2.Programming in PYTHON (ACTDSPPY001P)
3.Artificial Neural Networks (ACTDEANN001T/P)</td></tr>
<tr><td>Course Objectives</td><td colspan="7">1.Become an expert in neural networks
2.Learn to implement NNs in Keras and TensorFlow.
3.Build convolutional networks for image recognition.
4.Build recurrent networks for sequence generation.
5.Build generative adversarial networks for image generation.</td></tr>
<tr><td>Course Content</td><td colspan="7">List of Practical:

Module 1: TensorFlow
1.Installing TensorFlow on System (Windows/Mac/Linux)
2.Using Constants, Variables, and Placeholders of TensorFlow calculate the area of a circle
3.Use Run and Fetches Command of TensorFlow to find the Sum, Product, Absolute Difference, and Common Factors of Two Numbers

Module 2: Deep learning
1.Construct a Simple Neural Network (Perceptron) using Keras
2.Implement Various Activation Functions (ReLU, Tanh, Sigmoid)
on a perceptron using Randomly Generated Data
3. Write a program to calculate Shanon Entropy, Cross-Entropy, and KL Divergence

Module 3: Training Network
1.Write a program implementing Gradient Descent on a Perceptron
2.Change the values of the Learning Rate Delta and observe the trend in Error</td></tr>
<tr><td>Text Books</td><td colspan="7">T1: Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 2nd Edition, Pearson Education.
T2: Elaine Rich, Kevin Knight, Shivshankar B Nair, Artificial Intelligence, McGraw Hill, 3rd Edition.
T3: Elaine Rich, Kevin Knight, Artificial Intelligence, Tata McGraw Hill, 2nd Edition.</td></tr>
<tr><td>References</td><td colspan="7">R1: Lugar, .AI-Structures and Strategies for Complex Problem Solving., 4/e, 2002, Pearson Education.
R2: Nils J. Nilsson, Principles of Artificial Intelligence, Narosa Publication.
R3: Patrick H. Winston, Artificial Intelligence, 3rd edition, Pearson Education.
R4: Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Publication</td></tr>
<tr><td>Course Outcomes</td><td colspan="7">After completing this course, students will be able to:

CO1. Students will develop a basic understanding of the building blocks of AI as presented in terms of intelligent agents.
CO2.Students will be able to choose an appropriate problem-solving method and knowledge-representation scheme.
CO3.Students will develop an ability to analyze and formalize the problem (as a state space, graph, etc.) and select the appropriate search method.
CO4.Students will be able to develop/demonstrate/ build simple intelligent systems</td></tr>
</table>

	CO5.Students will be able to develop/demonstrate/ build various classical to y problems using different AI techniques.

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, ..., K_6)
CO1	1	1	1	1	2						1	1	1	1	**K_1, K_2**
CO2	2	1	1	1	1						1	2		1	**K_1, K_2**
CO3	2	2	1	1	1						1	2	1		**K_1, K_2, K_3**
CO4	3	2	2	1	2				1		2	2	1	2	**K_1, K_2, K_3, K_4**
CO5	3	3	2	1	2			1		1	2	2	2	2	**K_1, K_2, K_3, K_4**

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

K_6 => Create

SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL						
School Name: School of Engineering & Technology			**Department Name: Advance Computing**			
Recommended Programs: MTech. In Artificial Intelligence					Semester: II	
Course Name	**Advance Database Management Systems**		**Course Code**	**ET25MTAI217P**		
Credit Hours	**L**	**T**	**P**	**N**	**Total Credits**	1
	0	0	2	0		
Prerequisites	● Computability, Algorithms, and Complexity ● Mathematics for Computing ● Data structures and complexity					
Course Objectives	1. Understand the importance of DBMS 2. Learn how DBMS is better than traditional File Processing Systems. 3. Analyze the basic structure of the Database and recognize the different views of the database. 4. Understand and explain the terms like Transaction Processing and Concurrency Control. 5. Understand types of database failure and recovery					
Course Content	**List of Practical** **1.** Advanced SQL: Perform queries for DCL Commands and Locks **2.** Implement authorization, authentication, and privileges on database **3.** Perform queries to Create synonyms, sequences, and indexes **4.** Perform queries to Create, Alter and update views. **5.** PL / SQL and Triggers: Implement PL/SQL programs using control structures **6.** Implement PL/SQL programs using Cursors **7.** Implement PL/SQL programs using exception handling **8.** Implement user-defined procedures and functions using PL/SQL blocks **9.** Perform various operations on packages **10.** Implement various triggers. **11.** Functional Dependency and Decomposition: Practice on functional dependencies **12.** Normalization: Practice on Normalization- using any database to perform various normal forms. **13.** Transaction Processing: Practice transaction processing Cloud Databases: MongoDB/Cassandra etc.					
Text Books	T1: Henry F. Korth and Silberschatz Abraham, "Database System Concepts", Mc.Graw Hill. T2: Elmasri Ramez and Novathe Shamkant, "Fundamentals of Database Systems", Benjamin Cummings					
References	R1: James Martin, "Principles of Database Management Systems", 1985, Prentice Hall of India, New Delhi R2: "Fundamentals of Database Systems", Ramez Elmasri, Shamkant B.Navathe, Addison Wesley					
Course Outcomes	**After completing this course, students will be able to:** **CO1**: The students will be able to define, explain and illustrate the fundamental concepts of databases. **CO2**: The students will be able to model and design a relational database following the design principles **CO3**: The students will be able to define, explain and illustrate fundamental principles of data organization, query optimization, and concurrent transaction processing. **CO4**: The students will be able to appreciate the latest trends in databases. **CO5:** Understand types of database failure and recovery					

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	Knowledge Levels (K_1, K_2, ..., K_6)
CO1	1	1	1	1	2						1	1	1	1	**K_1, K_2**
CO2	2	1	1	1	1					1	1	2		1	**K_1, K_2**
CO3	2	2	1	1	1						1	2	1		**K_1, K_2, K_3**
CO4	3	2	2	1	2				1		2	2	1	2	**K_1, K_2, K_3, K_4**
CO5	3	3	2	1	2			1		1	2	2	2	2	**K_1, K_2, K_3, K_4**

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

K_6 => Create

<table>
<tr><td colspan="7">SANJEEV AGRAWAL GLOBAL EDUCATIONAL (SAGE) UNIVERSITY, BHOPAL</td></tr>
<tr><td colspan="3">School Name: School of Engineering & Technology</td><td colspan="4">Department Name: Advance Computing</td></tr>
<tr><td colspan="5">Recommended Programs: MTech. In Artificial Intelligence</td><td colspan="2">Semester: II</td></tr>
<tr><td>Course Name</td><td colspan="2">Java Scripting with AJAX</td><td>Course Code</td><td colspan="3">ET25MTAI218P</td></tr>
<tr><td rowspan="2">Credit Hours</td><td>L</td><td>T</td><td>P</td><td>N</td><td rowspan="2">Total Credits</td><td rowspan="2">2</td></tr>
<tr><td>0</td><td>0</td><td>4</td><td>0</td></tr>
<tr><td>Prerequisites</td><td colspan="6">• Programming in Java</td></tr>
<tr><td>Course Objectives</td><td colspan="6">1. To learn basic concepts of html5 and style sheets.
2. To develop java scripting programs in Java.
3. To develop web-based applications using AJAX.
4. To develop web-based applications using jQuery.
5. Develop web-based applications using AJAX.</td></tr>
<tr><td>Course Content</td><td colspan="6">List of Practical
1.Write, test and debug small applications Using HTML tags, input tags, input types.
2.Write, test and debug small applications with HTML5 Semantic Page Elements, inline semantic elements, media semantic elements.
3.Design a Page execute all three types of CSS
Internal CSS,
External CSS
Inline CSS
Multiple CSS
4.Write, test and debug small applications with Basic CSS.
5.Write, test and debug a form and implement javascript showing all the major form validations.
6.Write test and debug a JavaScript program illustrating the importance of Window Object Model.
7.Write test and debug a JavaScript program illustrating the Date and math Objects.
8.Write test and debug a jQuery program representing the use of hide (), show () and toggle () functions.
9.Write test and debug a program implementing jQuery fading methods.
10.Write test and debug a program implementing mouse and keyboard events.</td></tr>
<tr><td>Text Books</td><td colspan="6">T1: JavaScript: Definitive Guide, By: David Flanagan Publisher:O'Reilly Media, May 2011 Pages:1096 Print ISBN: 978-0-596-80552-4 ISBN 10: 0-596-80552-7 Ebook ISBN: 978-1-4493-0212--2 ISBN 10: 1-4493-0212-2. Multiple inexpensive options: Amazon.com, and Valor Books. - Google Play
T2: JavaScript Sixth Edition by Sasha Vodnik and Don Gosselin, Course Technology Incorporated, 2014, ISBN 1305078446 - online options
T3: JavaScript, Fifth Edition by Don Gosselin; Course Technology Incorporated, 2011, ISBN 0-538-74887-7, 180 Day E-Book</td></tr>
<tr><td>References</td><td colspan="6">R1: jQuery and JQuery UI, Visual Quickstart Guide by Jay Blanchard, Publisher: Peachpit Press; 1 edition (December 13, 2012), ISBN-10: 0321885147 PDF first 59 Pages.
R2: Developing Web Pages with jQuery, Gosselin Copyright 2013 ISBN 13: 978-1-4354-6079-9</td></tr>
<tr><td>Course Outcomes</td><td colspan="6">After completing this course, students will be able to:
CO1: Students who complete this course will be able compose basic JavaScript programs including data types,</td></tr>
</table>

	CO2: Demonstrate representation of control structures, functions operators events. CO3 Identify Debugging Students who complete course will be able to explain browser specific debugging tools. CO4: Demonstrate general fundamental debugging techniques to fix JavaScript errors. CO5: Demonstrate basic knowledge AJAX and Web Application.

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PSO 1	PSO 2	PSO 3	Knowledge Levels (K_1, K_2, ..., K_6)
CO1	3	1	2								1	1	1	1	K_3
CO2	3	2	3	1	3						1	2	1	1	K_3
CO3	2	3	3	1	3						1	2	2	2	K_3
CO4	2	2	1	2	2	1			1	1	2	2	3	1	K_3
CO5	3	2	2	3	2						2	2	3	2	K_3

High-3 Medium-2 Low-1

K_1 => Remember K_2 => Understand K_3 => Apply K_4 => Analyze K_5 => Evaluate

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